

What Information Matters? Graph Out-of-Distribution Detection via Tri-Component Information Decomposition

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Abstract

Graph neural networks are widely used for node classification, but they remain vulnerable to out-of-distribution (OOD) shifts in node features and graph structure. Prior work established that methods trained with standard supervised learning (SL) objectives tend to capture spurious signals from either features and/or structure, leaving the model fragile under distributional changes. To address this, we propose TIDE, a **novel and effective Tri-Component Information Decomposition** framework that **explicitly decomposes information** into *feature-specific*, *structure-specific* and *joint* components. TIDE aims to **preserve only** the label-relevant part of the joint information while **filtering out** spurious feature- and structure-specific information, thereby enhancing the separation between in-distribution (ID) and OOD nodes. Beyond the framework, we provide theoretical and empirical analyses showing that an information bottleneck objective is preferable to standard SL for graph OOD detection, with higher ID confidence and a greater entropy gap between ID and OOD data. Extensive experiments across seven datasets confirm the efficacy of TIDE, achieving up to a 34% improvement in FPR95 over strong baselines while maintaining competitive ID accuracy. Code is available at [🔗](#).

1. Introduction

Graph neural networks (GNNs) have become a dominant approach for node classification across real-world domains; however, they remain vulnerable to out-of-distribution (OOD) shifts in node features and/or graph structure, limiting reliable deployment in real-world applications (Li et al., 2022a; Wu et al., 2023b; Gui et al., 2022; Yang et al., 2022;

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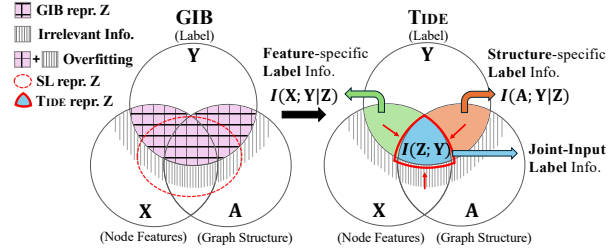


Figure 1. Comparison of information captured by standard supervised learning (SL), GIB (Wu et al., 2020), and TIDE. The dashed red circle shows that SL mixes label-relevant and irrelevant feature and structure signals, causing spurious correlations that make OOD nodes appear ID. The gridded region indicates that GIB suppresses label-irrelevant noise but still keeps input-specific spurious cues. Contrary, TIDE (solid red triangle) isolates the joint label-relevant information by decomposing spurious feature-/structure-specific components, thereby improving OOD detection.

Shen et al., 2024; Giuffrè & Shung, 2023). In biomedical graphs, node features like biomarkers may correlate with health outcomes in one hospital but shift in another, degrading performance despite stable patient-symptom links (**feature shift**). In e-commerce graphs, co-purchase links can change as shopping patterns evolve, even when product attributes remain stable (**structural shift**). In social networks, new communities may alter both posting behaviour and friendship links, causing simultaneous changes in features and structure signals (**joint shift**) (Wang et al., 2025b).

Given these challenges, OOD detection is critical for identifying nodes beyond the in-distribution (ID) training data (Lang et al., 2023; Zhou et al., 2022; Bazhenov et al., 2022). Existing graph OOD methods typically handle shifts by learning invariant node representations, using topology-aware metrics to capture node irregularities (Li et al., 2022b; Song & Wang, 2022; Bao et al., 2024), or applying post-hoc scoring functions to separate ID from OOD nodes (Wang et al., 2021; Liu et al., 2020; Wu et al., 2023b; Hendrycks & Gimpel, 2017; Hendrycks et al., 2022; Lee et al., 2018b). Others incorporate contrastive learning or OOD exposure when auxiliary OOD data is available (Hendrycks et al., 2019; Yang et al., 2024b).

While effective, many existing approaches rely on a **mixed representation of a graph’s features and structure**, without explicitly modelling the separate contributions of fea-

tures $\mathbf{X}$ and structure $\mathbf{A}$ (Wu et al., 2023b). This mixed representation corresponds to the node embeddings $\mathbf{Z} = f(\mathbf{X}, \mathbf{A})$ learned by a GNN f under a standard supervised-learning (SL) objective. As shown in Figure 1, this representation (red dotted circle) mixes label-irrelevant information (shaded area) with three label-related signals: the **feature-specific** signal that depends only on $\mathbf{X}$, the **structure-specific** signal from $\mathbf{A}$, and the **joint-input** signal arising from features-structure interactions. Although all three signals can be predictive in-distribution, prior work has shown that feature-only and structure-only signals often reflect spurious correlations with the label, which become problematic under distribution shifts (Chen et al., 2023; 2024; Fan et al., 2024). For example, under a feature shift with ID-like structure ($\mathbf{X}_{\text{OOD}}, \mathbf{A}_{\text{ID}}$), a model that exploits spurious structure-label correlations may confidently treat OOD nodes as ID. Similarly, under a structure shift with ID-like features ($\mathbf{X}_{\text{ID}}, \mathbf{A}_{\text{OOD}}$), reliance on feature-only correlations can again yield overconfident predictions that evade detection. In contrast, the joint-input signal is more shift-indicative because mismatches between features and structure are exposed during encoding. Thus, OOD detection can fail under SL when the model relies on spurious single-input correlations (from $\mathbf{X}$ or $\mathbf{A}$) rather than on joint-input information. A typical example with distribution visualisations is shown in Figure 2. When the dataset contains samples with ID features but OOD structure ($\mathbf{X}_{\text{ID}}, \mathbf{A}_{\text{OOD}}$), a standard SL-trained OOD detector (top left) fails to clearly separate OOD data from ID. Yet an expected behaviour from a successful detector should be a well separation given the structural shift as in bottom left or top right in Figure 2.

Ideally, to handle real-world distribution shifts, **an effective graph OOD detector should prioritise the label-relevant joint-input information while filtering out feature- and structure-specific spurious correlations**. Figure 1 illustrates this motivation: SL mixes feature-only, structure-only, and joint signals together with label-irrelevant information (red dotted circle); in contrast, compressing the learned representation toward the triangle-like overlap (red solid triangle) preserves primarily joint-input signals, which improves ID-OOD separation. To achieve this, we propose TIDE, a Tri-Component Information Decomposition framework. Unlike prior OOD detection methods that treat feature and structure information as a single unified signal, TIDE guides learning toward the desired joint-input region by explicitly decomposing predictive label information $I(\mathbf{X}, \mathbf{A}; \mathbf{Y})$ into: **feature-specific information** $I(\mathbf{X}; \mathbf{Y}|\mathbf{Z})$, **structure-specific information** $I(\mathbf{A}; \mathbf{Y}|\mathbf{Z})$, and **joint-input information** $I(\mathbf{Z}; \mathbf{Y})$. TIDE **preserves** the joint-input signal while suppressing individual-input components (indicated by the left and right arrows $\rightarrow$). Moreover, we employ an information bottleneck (IB) objective to filter out label-irrelevant noise (indicated by the upward arrow $\uparrow$), pro-

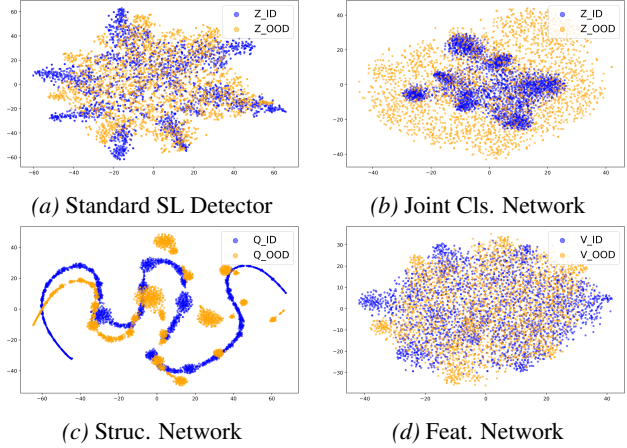


Figure 2. ID-OOD representation distributions for standard SL trained detector and TIDE’s joint classification (Cls.), structure (Struc.) and feature (Feat.) networks on Cora-Structure with a structure shift and ID-like features ($\mathbf{X}_{\text{ID}}, \mathbf{A}_{\text{OOD}}$).

ducing a compact and shift-indicative representation. We provide theoretical insights showing that, compared to SL, IB increases ID confidence and improves ID-OOD separation for logit-based detection. Thus, our **contributions** are:

- 1. Methodological:** We propose TIDE, a novel **tri-component information decomposition framework** for graph OOD detection that decomposes predictive information into **feature-specific**, **structure-specific**, and **joint** components, and introduces novel regularisers to suppress label-irrelevant and spurious correlations.
- 2. Theoretical:** We show that IB increases ID confidence and widens the entropy gap between ID and OOD relative to standard SL, improving logit-based OOD detection.
- 3. Empirical:** On seven real-world and synthetic datasets, TIDE achieves up to **34% improvement in FPR95** over baselines while maintaining competitive ID accuracy.

2. Related Work

Graph OOD detection builds on several lines of research.

1) Scoring-based methods rely on ID data to design OOD scores (Lee et al., 2018a; Koo et al., 2024; Ding & Shi, 2023; Ma et al., 2023; Liu et al., 2023; Hendrycks & Gimpel, 2017). **2) OOD exposure approaches** incorporate auxiliary OOD data during training for improved detection (Hendrycks et al., 2019; Liu et al., 2020; Park et al., 2023; Zhu et al., 2023; Yang et al., 2024b; Bao et al., 2024). **3) IB principle** has been applied to image and graph representation learning (Wu et al., 2020; Alemi et al., 2017; Tishby et al., 2000), but its role in OOD detection has mainly been studied in Euclidean domains (Zhao & Cao, 2023; Li et al., 2023b; Wu & Deng, 2024). Prior work analysed overconfidence only under the standard supervised objective for ID ($\max I(\mathbf{Z}; \mathbf{Y})$) (Hu et al., 2024), whereas our **theoretical contribution** (Sec. 5) characterises how the full IB objective improves ID confidence and logit-based OOD

detection – a gap unaddressed in earlier studies (Hu et al., 2024; Alemi et al., 2018). **4) Graph-specific methods** include GNNSAFE/++ with propagation-based detection (Wu et al., 2023b), NODESAFE/++ with constrained energy scores (Yang et al., 2024b), DeGEM with multi-hop energy-based modelling for heterophilic graphs (Chen et al., 2025), and GOLD with pseudo-OOD embedding synthesis (Wang et al., 2025a). Detailed review is in Appendix H.

3. Preliminary

Node Classification and Graph Representation. For node classification, a graph is represented as $\mathcal{G} = (\mathbf{X}, \mathbf{A})$, where $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the node feature matrix and $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the adjacency matrix, with number of nodes n and feature dimension d . Given C classes, each node i has a label $y_i \in \{1, 2, \dots, C\}$. In this paper, we consider two tasks:

Task 1: In-distribution Classification. Given test nodes from the same distribution as training ($P_{train}(\mathbf{X}, \mathbf{A}) = P_{test}(\mathbf{X}, \mathbf{A})$ and $P_{train}(\mathbf{y}|\mathbf{X}, \mathbf{A}) = P_{test}(\mathbf{y}|\mathbf{X}, \mathbf{A})$), the goal is to train an L -layer GNN to predict node labels $\hat{\mathbf{y}} \in \mathbb{R}^n$ (refer to Appendix D for further details on GNN):

$$\hat{\mathbf{y}} = \text{Softmax}(\text{GNN}(\mathbf{X}, \mathbf{A})).$$

Task 2: Out-of-distribution Detection. Here, the objective is to identify test nodes from a different distribution ($P_{train}(\mathbf{X}, \mathbf{A}) \neq P_{test}(\mathbf{X}, \mathbf{A})$ or $P_{train}(\mathbf{y}|\mathbf{X}, \mathbf{A}) \neq P_{test}(\mathbf{y}|\mathbf{X}, \mathbf{A})$). This is formulated by an OOD detector F with scoring function S and threshold τ :

$$F(\mathbf{x}, \mathbf{A}; \text{GNN}) = \begin{cases} \text{OOD}, & S(\mathbf{x}, \mathbf{A}; \text{GNN}) \geq \tau, \\ \text{ID}, & S(\mathbf{x}, \mathbf{A}; \text{GNN}) < \tau. \end{cases} \quad (1)$$

Extended definitions for structure, feature, joint, and label shifts are in Appendix I (Gui et al., 2022).

Energy-Based OOD Detection. The energy score is an effective scoring function for distinguishing OOD from ID samples (Liu et al., 2020). For a node i , the energy score is:

$$S(\mathbf{x}_i, \mathbf{A}; \text{GNN}) = e_i = -\log \sum_{c=0}^{C-1} \exp(\ell_{i,c}), \quad (2)$$

where $e_i \in \mathbb{R}$ indicates the energy score, $\ell_i \in \mathbb{R}^C$ are the logits from $\text{GNN}(\mathbf{X}, \mathbf{A})$. GNNSAFE (Wu et al., 2023b) adapts this to graphs via energy propagation:

$$\mathbf{e}^{(k)} = \alpha \mathbf{e}^{(k-1)} + (1 - \alpha) \mathbf{D}^{-1} \mathbf{A} \mathbf{e}^{(k-1)}, \quad (3)$$

where α controls propagation and $\mathbf{D}$ is the degree matrix. For OOD exposure, (Liu et al., 2020) further introduces energy regularisation to separate ID and OOD scores with thresholds t_{ID} and t_{OOD} :

$$\max \mathcal{L}_{\text{EReg}}, \text{ where } \mathcal{L}_{\text{EReg}} = \mathbb{E}_{i \sim P_{\text{ID}}} [\max(0, t_{\text{ID}} - e_i)]^2 + \mathbb{E}_{j \sim P_{\text{OOD}}} [\max(0, e_j - t_{\text{OOD}})]^2. \quad (4)$$

4. Method

Motivation. Graph data is inherently multi-modal with node features $\mathbf{X}$ and graph structure $\mathbf{A}$. However, the typical SL objective does not distinguish whether predictive signals arise jointly from $(\mathbf{X}, \mathbf{A})$ or from individual inputs. Shown in Figure 1, this mixing of information (dotted circle) allows SL-trained models to rely on feature- or structure-specific spurious correlations that may be predictive under ID but misleading under shifts (Li et al., 2023a). Thus, our aim is to learn the desired label-relevant joint signal while filtering out the label-irrelevant and spurious feature-/structure-only cues, as shown in Figure 1’s solid triangle region.

In light of this, we propose TIDE, a novel graph-tailored tri-component decomposition that separates label-relevant information into joint, feature-specific, and structure-specific components (Figure 3). The key idea is for the joint representation $\mathbf{Z}$ to capture stable feature-structure interactions, while auxiliary networks isolate spurious feature-only $\mathbf{V}$ and structure-only $\mathbf{Q}$ signals. In the following sections, we present our **novel tri-component information decomposition paradigm** (Section 4.1) and propose the **TIDE framework** (Section 4.2) for **practical optimisation**.

4.1. Information Based Decomposition Paradigm

To begin, we formalise the relationship between inputs, encoded representation, and labels as follows.

Proposition 4.1. *Given a representation $\mathbf{Z}$ encoded from inputs $(\mathbf{X}, \mathbf{A})$ by a network f parametrised by θ , and is maximised for predicting label $\mathbf{Y}$, we have $I(\mathbf{Z}; \mathbf{Y}|\mathbf{X}, \mathbf{A}) = 0$ and $I(\mathbf{X}, \mathbf{A}; \mathbf{Y}) = I(\mathbf{X}, \mathbf{A}, \mathbf{Z}; \mathbf{Y})$.*

Here, $I(\mathbf{X}, \mathbf{A}; \mathbf{Y})$ represents the maximum information that we have about the prediction of $\mathbf{Y}$ given inputs $(\mathbf{X}, \mathbf{A})$. Proposition 4.1 thus states that $\mathbf{Z}$ cannot contain more information about $\mathbf{Y}$ than is already present in $(\mathbf{X}, \mathbf{A})$. The proof is in Appendix C.1. This leads to the decomposition:

$$I(\mathbf{X}, \mathbf{A}; \mathbf{Y}) = I(\mathbf{X}, \mathbf{A}, \mathbf{Z}; \mathbf{Y}) = \overset{\text{Chain rule for mutual information}}{I(\mathbf{Z}; \mathbf{Y})} + \underbrace{I(\mathbf{X}, \mathbf{A}; \mathbf{Y}|\mathbf{Z})}_{\text{residual}}, \quad (5)$$

where the last term captures label information not encoded by $\mathbf{Z}$. To isolate input-specific signals, this residual in Eq. 5 is decomposed into feature and structure components. (6)

$$I(\mathbf{X}, \mathbf{A}; \mathbf{Y}) = I(\mathbf{Z}; \mathbf{Y}) + \underbrace{I(\mathbf{X}; \mathbf{Y}|\mathbf{Z}) + I(\mathbf{A}; \mathbf{Y}|\mathbf{Z}, \mathbf{X})}_{\text{residual}}.$$

To effectively decouple $\mathbf{X}$ and $\mathbf{A}$, we enforce a conditional independence constraint $\mathbf{A} \perp\!\!\!\perp \mathbf{X} \mid \mathbf{Z}$, which is not a restrictive dataset assumption but a modelling objective. More discussions is provided in Sec 4.2. This constraint guides $\mathbf{Z}$ to retain all label-relevant joint information while reducing residual dependence between $\mathbf{X}$ and $\mathbf{A}$:

$$p(\mathbf{X}, \mathbf{A}|\mathbf{Z}) = p(\mathbf{X}|\mathbf{Z})p(\mathbf{A}|\mathbf{Z}). \quad (7)$$

label-relevant information while reducing label-irrelevant noise that blurs the ID-OOD boundary (details on IB are in Appendix E). This aligns with our decomposition (Eq. 8, Eq. 9): $\mathbf{Z}$ captures the joint-input label information, while $\mathbf{V}$ and $\mathbf{Q}$ capture the residual feature- and structure-specific components. By reducing spurious input correlations, IB strengthens OOD detection over standard SL, with theoretical analysis detailed in Section 5, Proposition 5.3.

$$\begin{aligned} \text{IB}_{\mathbf{Z}} &= \max I(\mathbf{Z}; \mathbf{Y}) - \beta_{\mathbf{Z}} I(\mathbf{X}, \mathbf{A}; \mathbf{Z}) \\ \text{IB}_{\mathbf{V}} &= \max I(\mathbf{V}; \mathbf{Y}) - \beta_{\mathbf{V}} I(\mathbf{X}; \mathbf{V}), \\ \text{IB}_{\mathbf{Q}} &= \max I(\mathbf{Q}; \mathbf{Y}) - \beta_{\mathbf{Q}} I(\mathbf{A}; \mathbf{Q}) \\ \max \mathcal{L}_{\text{IB}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}}, \text{ where } \mathcal{L}_{\text{IB}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}} &= \text{IB}_{\mathbf{Z}} + \text{IB}_{\mathbf{V}} + \text{IB}_{\mathbf{Q}}, \end{aligned} \quad (12)$$

Final Objective. The final TIDE objective combines these terms and regularisers to realise the objective in Eq. 9.

$$\begin{aligned} \max_{\theta_{\mathbf{Z}}, \theta_{\mathbf{V}}, \theta_{\mathbf{Q}}} \mathcal{L}_{\text{TIDE}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}} &= \\ \max_{\theta_{\mathbf{Z}}, \theta_{\mathbf{V}}, \theta_{\mathbf{Q}}} \mathcal{L}_{\text{IB}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}} - \lambda_{\text{CInd}} \mathcal{L}_{\text{CInd}_{\mathbf{X}, \mathbf{A}, \mathbf{Z}}} - \mathcal{L}_{\text{PMI}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}}, \end{aligned} \quad (13)$$

where λ_{CInd} is a loss coefficient and the scale weight for $\mathcal{L}_{\text{PMI}}$ is handled by $\alpha_{1,2,3}$ in Eq. 11.

We highlight that the final objective components are complementary rather than conflicting. The IB terms encourage each network to retain label-relevant information from its own inputs while discarding label-irrelevant noise. The PMI regulariser does not suppress task-relevant content; instead, it penalises redundant encoding of the same information, regulating where predictive information is learned. Thus, shared label-relevant feature-structure interactions are captured in $\mathbf{Z}$, while $\mathbf{V}$ and $\mathbf{Q}$ encode the remaining complementary signals. Moreover, the compression term for $\mathbf{Z}$ can be rewritten as $(1 - \beta_{\mathbf{Z}})I(\mathbf{Z}; \mathbf{Y}) - \beta_{\mathbf{Z}}I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y})$, which for a moderate $\beta_{\mathbf{Z}}$, preserves joint information that contributes to prediction rather than eliminating it. $\mathcal{L}_{\text{CInd}}$ is then applied to suppress residual feature-structure correlations that are not explained by $\mathbf{Z}$. Together, this formulation ensures $\mathbf{Z}$ provides a stable classification and OOD detection backbone, while $\mathbf{V}$ and $\mathbf{Q}$ act only as training regularisers, guiding disentanglement without affecting inference.

4.3. Practical Implementation and Inference

We note that **direct computation of mutual information in the final objective (Eq. 13) is intractable**, so we adopt standard variational approximations. IB terms are estimated via VIB (Alemi et al., 2017), while reconstruction loss and CLUB (Cheng et al., 2020) are used for the conditional independence and PMI regularisers. Appendix K discusses alternative MI estimators. Due to the page limit, and *to emphasise our primary contribution*, the detailed discussion on obtaining the **tractable objective** is instead provided in Appendix F, and a **neural network parameterisation** of TIDE is given in Appendix G. Each encoder is

Algorithm 1 TIDE Framework

Input: ID graph $\mathcal{G} = (\mathbf{X}, \mathbf{A})$, randomly initialised GNN classifier ($\text{GNN}_{\text{CLS-Z}}$), MLP feature network ($\text{MLP}_{\text{Feat-V}}$), structure network ($\text{GNN}_{\text{Struct-Q}}$) with parameters $\theta_{\mathbf{Z}}, \theta_{\mathbf{V}}, \theta_{\mathbf{Q}}$ respectively, loss coefficients $\alpha_1, \alpha_2, \alpha_3, \lambda_{\text{CInd}}$.

Output: Optimised $\text{GNN}_{\text{CLS-Z}}$.

while *train* **do**

Update:

1. Minimising $-\mathcal{L}_{\text{VIB}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}}$ **Eq. 38**

$\text{GNN}_{\text{CLS-Z}} \leftarrow -\text{VIB}_{\mathbf{Z}}$

$\text{MLP}_{\text{Feat-V}} \leftarrow -\text{VIB}_{\mathbf{V}}$

$\text{GNN}_{\text{Struct-Q}} \leftarrow -\text{VIB}_{\mathbf{Q}}$

2. Minimising $\mathcal{L}_{\text{VCInd}_{\mathbf{X}, \mathbf{A}, \mathbf{Z}}}$ **Eq. 39**

$\text{GNN}_{\text{CLS-Z}} \leftarrow \lambda_{\text{CInd}} \mathcal{L}_{\text{VCInd}_{\mathbf{X}, \mathbf{A}, \mathbf{Z}}}$

3. Minimising $\mathcal{L}_{\text{VPMI}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}}$ **Eq. 41**

$\text{GNN}_{\text{CLS-Z}} \leftarrow \alpha_1 I_{\text{VCLUB}_{\mathbf{Z}, \mathbf{V}}} + \alpha_2 I_{\text{VCLUB}_{\mathbf{Z}, \mathbf{Q}}}$

$\text{MLP}_{\text{Feat-V}} \leftarrow \alpha_1 I_{\text{VCLUB}_{\mathbf{Z}, \mathbf{V}}} + \alpha_3 I_{\text{VCLUB}_{\mathbf{V}, \mathbf{Q}}}$

$\text{GNN}_{\text{Struct-Q}} \leftarrow \alpha_2 I_{\text{VCLUB}_{\mathbf{Z}, \mathbf{Q}}} + \alpha_3 I_{\text{VCLUB}_{\mathbf{V}, \mathbf{Q}}}$

end while

Algorithm 2 TIDE Inference

- 1: **Input:** Test graph $\mathcal{G} = (\mathbf{X}, \mathbf{A})$, optimised GNN classifier ($\text{GNN}_{\text{CLS-Z}}$) with parameter $\theta_{\mathbf{Z}}$.
 - 2: **Output:** Predicted labels and energy scores.
 - 3: **1. Inference: Obtain logits**
 - 4: $\ell \leftarrow \text{GNN}_{\text{CLS-Z}}(\mathcal{G})$
 - 5: **2. Score calculation**
 - 6: Energy: $\mathbf{e} \leftarrow$ Eq. 2, 3 using ℓ .
 - 7: Prediction: $\hat{\mathbf{Y}} \leftarrow \text{Softmax}(\ell)$.
-

trained only with losses corresponding to its intended information component: the $\mathbf{Z}$ network is optimised by the $\text{IB}_{\mathbf{Z}}$ term and regularisers to capture joint-input signals, whereas the $\mathbf{V}$ and $\mathbf{Q}$ networks are trained with their own IB objectives and pairwise MI penalties to isolate feature- and structure-specific signals. Without loss of generality, in the following sections, we refer to the mutual information and TIDE objective as their tractable variational forms (i.e., $\mathcal{L}_{\text{VIB}}, \mathcal{L}_{\text{VCInd}}, \mathcal{L}_{\text{VPMI}}$). Algorithm 1 outlines the full optimisation loop, including the forward pass through all modules, computation of VIB, CLUB, and reconstruction terms, and joint gradient updates for all parameters of TIDE. **At inference time** (Algorithm. 2), only the GNN classifier for $\mathbf{Z}$ will be used for OOD detection with the energy score in Eq. 3 derived from prediction logits. Auxiliary networks $\mathbf{V}$ and $\mathbf{Q}$ act solely as training-time regularisers and do not participate in inference.

5. How Does IB Benefit Graph OOD Detection

Building on our novel decomposition framework, we further demonstrate how IB benefits graph OOD detection over

standard SL. Unlike SL ($\max I(\mathbf{Z}; \mathbf{Y})$), which rewards any predictive correlation, IB retains only label-supporting information, yielding **two key benefits**: (i) sharper ID prediction confidence and (ii) a larger ID-OOD entropy gap, which **directly improves logit-based OOD detection**.

Lemma 5.1 (Target-Irrelevant Information and ID Prediction Confidence). *Minimising the conditional mutual information $I(\mathbf{X}, \mathbf{A}; \mathbf{Z} \mid \mathbf{Y})$ reduces conditional entropy $H(\mathbf{Z} \mid \mathbf{Y})$, yielding a more concentrated posterior $P(\mathbf{Y} \mid \mathbf{Z})$.*

Lemma 5.1 shows that when the representation $\mathbf{Z}$ discards target-irrelevant information, the uncertainty of $\mathbf{Z}$ conditioned on $\mathbf{Y}$ decreases. In other words, the representation becomes more predictable given the label, which sharpens the posterior distribution $P(\mathbf{Y} \mid \mathbf{Z})$. Intuitively, as $H(\mathbf{Z} \mid \mathbf{Y})$ approaches zero, the peaks in the posterior become sharper, meaning the model assigns higher maximum probability to the correct class. This provides the first link between minimising irrelevant information and achieving higher confidence on ID predictions. Proof is shown in Appendix C.2.

Theorem 5.2 (ID Confidence Improvement for IB over SL). *Let $\mathbf{Z} = f(\mathbf{X}, \mathbf{A})$ be an encoded representation of $(\mathbf{X}, \mathbf{A})$. With information bottleneck: $\max I(\mathbf{Z}; \mathbf{Y}) - \beta I(\mathbf{X}, \mathbf{A}; \mathbf{Z})$, where $\beta \in (0, 1)$, the model attains higher in-distribution prediction confidence compared to standard supervised learning ($\beta = 0$), provided $I(\mathbf{X}, \mathbf{A}; \mathbf{Z} \mid \mathbf{Y})$ is minimised.*

Theorem 5.2 builds directly on Lemma 5.1. By rewriting the IB objective, we obtain:

$$\max(1 - \beta)I(\mathbf{Z}; \mathbf{Y}) - \beta I(\mathbf{X}, \mathbf{A}; \mathbf{Z} \mid \mathbf{Y}). \quad (14)$$

Theorem 5.2 therefore guarantees that models trained with IB achieve higher prediction confidence on ID data than standard SL (e.g., $\beta = 0$). Proof is provided in Appendix C.3.

Proposition 5.3 (IB Objective Increases Entropy Separation between ID and OOD). *Let $\mathbf{Z}^*$ be the representation obtained from an optimal network trained with ID data via the IB objective (Eq. 36). Then:*

1. $I(\mathbf{X}_{id}, \mathbf{A}_{id}; \mathbf{Z}_{id}^* \mid \mathbf{Y}) \rightarrow 0$,
2. $I(\mathcal{G}_{OOD}; \mathbf{Z}_{ood}^* \mid \mathbf{Y}) \geq I(\mathbf{X}_{id}, \mathbf{A}_{id}; \mathbf{Z}_{id}^* \mid \mathbf{Y})$.

This induces entropy separation:

$$H(\mathbf{Y} \mid \mathbf{Z}_{id}^*) \ll H(\mathbf{Y} \mid \mathbf{Z}_{ood}^*),$$

enabling improved OOD detection via logit-based scores.

Proposition 5.3 formalises how the advantages of IB for ID confidence translate into OOD detection benefits. The proof is provided in Appendix C.4. For ID data, point (1) follows from Theorem 5.2: the conditional mutual information vanishes, which yields low entropy predictions and sharp confidence. For OOD data, point (2) shows that the

information compressed into $\mathbf{Z}$ inevitably contains more irrelevant content, resulting in higher conditional entropy and lower confidence. The contrast between these two cases induces a larger entropy gap between ID and OOD data under IB training than under SL training. This entropy separation directly improves logit-based OOD scores such as energy, making IB-trained models naturally better suited for OOD detection. Empirical evaluations in Section 6 validate this theoretical analysis for OOD detection.

6. Experiments

Datasets & Metrics. Following (Wu et al., 2023b; Yang et al., 2024b), we evaluate on seven benchmarks. Six are single-graph datasets (Cora, Citeseer, Pubmed, Amazon-Photo, Coauthor-CS, ogbn-Arxiv), with OOD nodes generated via structure manipulation, feature interpolation, label exclusion, or temporal splits. We also use the multi-graph TwitchGamers-Explicit, where OOD corresponds to graphs from different regions. Performance is measured by AUROC ($\uparrow$), AUPR ($\uparrow$), and FPR95 ($\downarrow$) for OOD detection, along with ID accuracy (ID ACC) ($\uparrow$). Details on datasets/metrics are in Appendix L/J.

Baselines. To fairly evaluate our method, we select **SOTA baselines trained with standard SL**: General OOD methods: MSP (Hendrycks & Gimpel, 2017), ODIN (Liang et al., 2018), Mahalanobis (Lee et al., 2018b), Energy (Liu et al., 2020). Graph-specific methods: GKDE (Zhao et al., 2020), GPN (Stadler et al., 2021), GNNSAFE (Wu et al., 2023b), NODESAFE (Yang et al., 2024b). OOD exposure method: OE (Hendrycks et al., 2019), Energy FT (Liu et al., 2020), GNNSAFE++, NODESAFE++. We further evaluate two advanced graph OOD detection methods, DeGEM (Chen et al., 2025) and GOLD (Wang et al., 2025a), in Table 5.

Implementation. All methods use a two-layer GCN backbone (hidden size 64). The feature and structure networks are implemented as an MLP and a GCN, respectively, with the same architecture. For the structure network, node features are fixed to 1, while the feature network uses only raw node attributes. Additional setups, design choices, hyperparameters, and sensitivity analysis are given in Appendix K.

6.1. Overall Performance

As a non-OOD exposed method, TIDE markedly outperforms SOTA non-OOD exposure baselines. Evident in Table 1, TIDE significantly enhances OOD detection performance across all datasets (blue highlights). On synthetic datasets such as Citeseer and Pubmed, TIDE reduces FPR95 by an average of 24%. While datasets like Amazon and Coauthor exhibit strong performance with existing SOTA baselines, driven by high classification accuracy, which yield more discriminative energy scores for OOD

Table 1. Model performance with Non-OOD exposure. Following (Wu et al., 2023b; Yang et al., 2024b), the results are **averaged over multiple OOD test sets with different difficulty levels**, therefore with a relatively high variance $\pm$ across subsets. Individual subset results with a much lower variance are in Appendix M. The best and runner-up results are highlighted by **best** and **runner-up**, respectively.

	Metrics	MSP	ODIN	Maha	Energy	GKDE	GPN	GNNSAFE	NODESAFE	TIDE
Cora	AUROC ($\uparrow$)	82.55	49.87	54.74	83.09	69.54	84.56	91.20 $\pm$ 3.09	93.35 $\pm$ 2.41	95.58 $\pm$ 2.12
	AUPR ($\uparrow$)	65.82	26.08	34.43	66.21	46.09	68.02	82.92 $\pm$ 4.96	84.64 $\pm$ 6.40	89.34 $\pm$ 5.32
	FPR95 ($\downarrow$)	62.39	100.00	96.30	65.21	80.51	58.30	50.53 $\pm$ 22.04	29.23 $\pm$ 12.06	20.09 $\pm$ 10.72
	ID ACC ($\uparrow$)	79.91	79.61	79.57	80.34	79.86	81.65	81.56 $\pm$ 7.01	82.66 $\pm$ 6.61	82.56 $\pm$ 6.99
Citeseer	AUROC ($\uparrow$)	77.69	50.14	49.55	78.26	72.95	78.22	84.89 $\pm$ 5.47	87.80 $\pm$ 2.74	92.27 $\pm$ 1.53
	AUPR ($\uparrow$)	51.43	21.38	29.29	51.22	47.67	52.22	64.86 $\pm$ 3.39	72.35 $\pm$ 6.41	76.57 $\pm$ 10.34
	FPR95 ($\downarrow$)	69.42	100	95.06	65.34	71.85	67.09	60.85 $\pm$ 18.06	53.38 $\pm$ 17.47	30.79 $\pm$ 7.64
	ID ACC ($\uparrow$)	73.72	73.75	62.17	73.43	72.69	72.77	76.06 $\pm$ 12.05	73.12 $\pm$ 14.15	75.66 $\pm$ 11.19
Pubmed	AUROC ($\uparrow$)	78.80	49.72	62.20	79.25	78.14	78.76	93.82 $\pm$ 1.93	91.08 $\pm$ 2.95	97.35 $\pm$ 0.13
	AUPR ($\uparrow$)	28.37	4.83	11.74	28.21	24.65	28.65	69.94 $\pm$ 6.32	68.56 $\pm$ 1.06	80.73 $\pm$ 1.37
	FPR95 ($\downarrow$)	76.73	100	91.26	70.69	75.04	71.06	37.71 $\pm$ 14.28	50.17 $\pm$ 0.89	12.75 $\pm$ 0.31
	ID ACC ($\uparrow$)	75.05	75.30	71.15	75.55	74.65	75.15	76.78 $\pm$ 0.31	77.32 $\pm$ 0.02	76.17 $\pm$ 0.33
Amazon	AUROC ($\uparrow$)	96.52	80.12	73.81	96.73	66.98	92.60	97.99 $\pm$ 0.93	97.84 $\pm$ 1.11	98.16 $\pm$ 1.14
	AUPR ($\uparrow$)	95.01	77.18	72.35	95.16	71.18	90.50	98.16 $\pm$ 1.56	97.77 $\pm$ 2.11	98.08 $\pm$ 1.99
	FPR95 ($\downarrow$)	13.83	85.22	83.44	13.15	98.47	32.64	3.24 $\pm$ 5.24	3.69 $\pm$ 6.14	2.81 $\pm$ 4.61
	ID ACC ($\uparrow$)	93.83	93.88	93.80	93.85	87.71	89.54	93.79 $\pm$ 1.99	92.70 $\pm$ 2.16	93.90 $\pm$ 1.88
Coauthor	AUROC ($\uparrow$)	95.74	51.71	82.02	96.64	69.24	69.89	98.98 $\pm$ 1.41	99.02 $\pm$ 1.45	99.27 $\pm$ 1.17
	AUPR ($\uparrow$)	96.43	56.37	87.05	97.09	80.17	72.77	99.55 $\pm$ 0.44	99.57 $\pm$ 0.47	99.70 $\pm$ 0.40
	FPR95 ($\downarrow$)	21.37	99.97	48.09	15.49	97.04	69.60	4.29 $\pm$ 6.87	4.33 $\pm$ 7.04	3.31 $\pm$ 5.45
	ID ACC ($\uparrow$)	93.37	93.29	93.29	93.57	87.74	89.39	93.65 $\pm$ 1.50	93.21 $\pm$ 1.86	93.48 $\pm$ 1.74
Twitch	AUROC ($\uparrow$)	33.59	58.16	55.68	51.24	46.48	51.73	66.33 $\pm$ 15.32	66.48 $\pm$ 15.44	89.72 $\pm$ 5.42
	AUPR ($\uparrow$)	49.14	72.12	66.42	60.81	62.11	66.36	72.59 $\pm$ 13.44	72.71 $\pm$ 13.43	91.92 $\pm$ 3.85
	FPR95 ($\downarrow$)	97.45	93.96	90.13	91.61	95.62	95.51	81.18 $\pm$ 19.87	80.62 $\pm$ 20.03	46.60 $\pm$ 26.47
	ID ACC ($\uparrow$)	68.72	70.79	70.51	70.40	67.44	68.09	70.75 $\pm$ 0.30	70.75 $\pm$ 0.33	68.63 $\pm$ 1.41
Arxiv	AUROC ($\uparrow$)	63.91	55.07	56.92	64.20	58.32	OOM	70.58 $\pm$ 6.41	71.36 $\pm$ 6.35	72.72 $\pm$ 6.48
	AUPR ($\uparrow$)	75.85	68.85	69.63	75.78	72.62	OOM	79.99 $\pm$ 12.80	80.67 $\pm$ 12.37	81.62 $\pm$ 11.86
	FPR95 ($\downarrow$)	90.59	100.0	94.24	90.80	93.84	OOM	87.90 $\pm$ 2.57	86.45 $\pm$ 2.97	83.65 $\pm$ 4.14
	ID ACC ($\uparrow$)	53.78	51.39	51.59	53.36	50.76	OOM	53.21 $\pm$ 0.25	53.10 $\pm$ 0.21	52.44 $\pm$ 0.24

detection - TIDE further improves detection performance. This shows the efficacy of our dedicated information learning and IB over the standard SL in GNNSAFE. For real-world datasets, TIDE delivers remarkable improvements, increasing the AUROC score by over 23% and reducing the FPR95 by 34% on the Twitch dataset. On the more challenging Arxiv dataset, where performance is constrained by limited classification accuracy, TIDE still outperforms baseline methods. This highlights TIDE’s superiority in OOD detection while maintaining strong ID classification accuracy. Extended results are in Appendix M. Moreover, TIDE achieves **similar efficiency in inference speed and memory usage** as SOTA methods shown in Appendix N.

6.2. IB vs. SL in OOD Detection

The empirical results strongly validate IB’s superiority over SL for OOD detection. Table 2 compares SL-based and IB-based classifiers, with energy as the OOD score. Even in its simplest form, IB-based models consistently outperform their SL counterparts ($SL < IB$), highlighting IB’s fundamental advantages for OOD detection. When enhanced with effective energy propagation techniques, the

Table 2. Detection comparison between IB vs. SL.

	Metrics	w/o energy propagation		w/ energy propagation	
		SL	IB	SL	IB
Cora	AUROC ($\uparrow$)	83.09	86.90	91.20	95.18
	AUPR ($\uparrow$)	66.21	69.89	82.92	88.49
	FPR95 ($\downarrow$)	65.21	46.26	50.53	22.20
Pubmed	AUROC ($\uparrow$)	79.25	87.04	84.49	91.50
	AUPR ($\uparrow$)	29.21	45.87	64.86	75.22
	FPR95 ($\downarrow$)	70.69	52.73	60.85	37.42
Twitch	AUROC ($\uparrow$)	51.24	70.59	66.33	73.50
	AUPR ($\uparrow$)	60.81	73.26	72.59	76.67
	FPR95 ($\downarrow$)	91.61	82.46	81.18	67.26

performance gap widens further, with IB models showing significantly greater improvements.

6.3. Visualisations of TIDE

Energy Gap. To validate our information-decomposed framework and highlight the advantage of IB over the SL baseline for OOD detection, Figure 4 visualises the **prediction confidence and energy distributions** of TIDE and GNNSAFE. The first subplot supports the theoretical insight that the IB objective enhances ID prediction confidence: the **blue distribution** of TIDE is skewed toward higher confidence values than the **green distribution** of GNNSAFE.

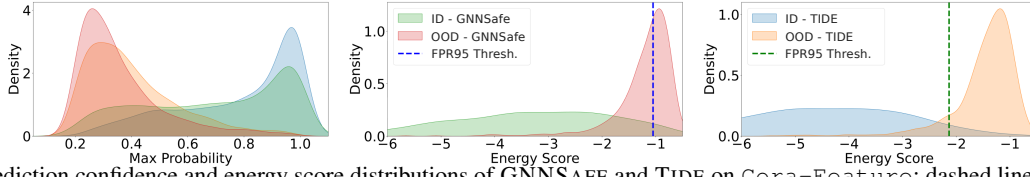


Figure 4. Prediction confidence and energy score distributions of GNNSAFE and TIDE on Cora-Feature; dashed line shows FPR95.

Additionally, the second and third subplot shows that **TIDE achieves a greater separation between ID and OOD energy scores than GNNSAFE**. This increased energy margin highlights TIDE’s efficacy in distinguishing ID from OOD data. The dashed FPR95 thresholds indicate reduced ID-OOD overlap for TIDE, explaining its stronger OOD detection while maintaining ID accuracy.

Connecting this visualisation to theory, Proposition 5.3 shows that the IB objective increases ID confidence and enlarges the ID-OOD entropy gap compared to standard SL. This analysis is stated in terms of the predictive distribution $p_\theta(\mathbf{y} | \mathbf{z}) = \text{Softmax}(\ell)$ and its entropy $H(\mathbf{Y} | \mathbf{Z}) = -\sum_y p_\theta(y | \mathbf{z}) \log p_\theta(y | \mathbf{z})$, where ℓ are the node logits. Since the energy score (Eq. 2) is computed from the same logits and depends on the same log-partition term in the entropy, IB-induced sharpening of ID posteriors (lower entropy) and increased uncertainty on OOD inputs (higher entropy) translate into lower energies for ID nodes and higher energies for OOD nodes, as shown in Figure 4.

Representation Distribution. Under the tri-component decomposition, **TIDE learns more shift-indicative representations**, particularly via the joint-input network. Figure 2 compares ID and OOD representations from (1) a **standard SL-trained GCN** and from the (2) **joint**, (3) **structure**, and (4) **feature** networks of TIDE. Notably, the joint network achieves clearer ID-OOD separation than the baseline GCN. On the structure-shifted dataset (Cora-S), the feature network exhibits little separation, while the structure network captures a distinct, separable topology pattern.

6.4. Ablation Study

We conduct an ablation study on the IB backbone, conditional independence constraint, and the full TIDE framework, which includes a pairwise mutual information minimisation loss to reduce spurious correlations. Compared to GNNSAFE (standard SL trained), the IB backbone lowers FPR95 by 20% on average, highlighting the benefit of compressing irrelevant features. Adding the independence constraint that improves joint-input modelling yields a further 17% FPR95 reduction on Twitch, which exhibits a higher correlation between structure and feature than the synthetically created data. The complete TIDE performs best overall, learning robust representations for both ID classification and OOD detection. Details are in Appendix M.

Table 3. Ablation study.

	Metrics	Cora	Citeseer	Pubmed	Twitch	Arxiv
GNNSAFE ($\mathcal{L}_{\text{sup}}$)	AUROC ($\uparrow$)	91.20	84.89	93.82	66.33	70.58
	AUPR ($\uparrow$)	82.92	64.86	69.94	72.59	79.99
	FPR95 ($\downarrow$)	50.53	60.85	37.71	81.18	87.90
	ID ACC ($\uparrow$)	81.56	76.06	76.78	70.75	53.21
$\mathcal{L}_{\text{vib}}$	AUROC ($\uparrow$)	95.18	91.50	96.58	73.50	72.15
	AUPR ($\uparrow$)	88.49	75.22	77.34	76.67	81.13
	FPR95 ($\downarrow$)	22.20	37.42	16.85	67.26	84.93
	ID ACC ($\uparrow$)	82.15	75.39	74.95	70.36	52.46
$\mathcal{L}_{\text{vib}} \& \mathcal{L}_{\text{vib}}^{\text{ind}}$	AUROC ($\uparrow$)	95.23	91.99	96.94	87.61	72.68
	AUPR ($\uparrow$)	88.65	76.01	78.77	90.61	81.57
	FPR95 ($\downarrow$)	21.56	32.57	14.37	49.91	83.86
	ID ACC ($\uparrow$)	82.12	75.24	75.18	70.33	52.43
TIDE	AUROC ($\uparrow$)	95.58	92.27	97.35	89.72	72.72
	AUPR ($\uparrow$)	89.34	76.57	80.73	91.92	81.62
	FPR95 ($\downarrow$)	20.09	30.79	12.75	46.60	83.65
	ID ACC ($\uparrow$)	82.56	75.66	76.17	68.63	52.44

6.5. TIDE against Disentanglement-Based Methods

Table 4 compares TIDE with DeGEM (Chen et al., 2025) and GraphIFE (Zeng et al., 2025). DeGEM is a specialised graph OOD detector and therefore achieves competitive performance. For a fairer comparison, we also report a TIDE-enhanced DeGEM variant, which consistently improves logit-based detection metrics, supporting our theoretical finding that the IB principle sharpens ID predictions and increases ID-OOD separation. In contrast, GraphIFE is not designed for node-level OOD detection and shows limited energy-based detection capability across all datasets, even with energy propagation, despite reasonable ID accuracy.

Table 4. Comparison with disentanglement-based method.

Dataset	Method	AUROC	AUPR	FPR95	IDAcc
Cora-S	TIDE	95.15	89.33	23.31	78.97
	DeGEM	92.02	70.24	18.91	79.80
	DeGEM + TIDE	96.43	90.18	15.03	77.80
	GraphIFE	72.70	51.68	84.68	78.90
	GraphIFE + Prop	83.03	75.67	77.70	–
Cora-F	TIDE	97.88	94.67	8.13	78.10
	DeGEM	98.66	96.14	5.76	82.80
	DeGEM + TIDE	99.05	96.86	3.62	82.20
	GraphIFE	80.45	70.74	78.84	78.75
	GraphIFE + Prop	84.83	75.39	73.14	–
Pubmed-S	TIDE	97.25	79.76	12.97	75.93
	DeGEM	95.37	51.55	20.07	73.10
	DeGEM + TIDE	97.20	55.60	8.44	78.40
	GraphIFE	67.88	17.93	99.84	76.00
	GraphIFE + Prop	75.35	47.39	97.55	–
Pubmed-F	TIDE	97.44	81.70	12.53	76.40
	DeGEM	99.63	93.40	1.70	79.00
	DeGEM + TIDE	99.67	91.73	0.59	78.70
	GraphIFE	71.02	31.68	99.97	75.95
	GraphIFE + Prop	74.32	42.27	89.98	–

Table 5. Comparison of TIDE with advanced graph OOD detection methods. +TIDE denotes incorporating the TIDE objective. We mark an improved performance from integrating TIDE into the method in **Green**.

Model	Cora – Label				Pubmed – S				Pubmed – F				Amazon – F				Citeseer – S			
	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc
GNNSAFE	93.19	82.99	29.55	89.66	92.45	65.47	47.81	77.00	95.18	74.41	27.62	76.57	98.46	98.90	0.44	92.65	79.87	61.44	74.45	65.20
TIDE	93.70	84.03	28.84	90.61	97.25	79.76	12.97	75.93	97.44	81.70	12.53	76.40	98.65	99.04	0.30	92.70	91.89	80.11	38.41	70.03
DeGEM	92.24	78.80	31.34	91.77	95.37	51.55	20.07	73.10	99.63	93.40	1.70	79.00	97.34	96.70	3.16	91.65	94.93	84.63	17.49	70.90
+ TIDE	95.85	89.24	20.59	92.41	97.20	55.60	8.44	78.40	99.67	91.73	0.59	78.70	97.71	96.49	1.84	92.32	96.92	85.63	7.24	69.70
GOLD	95.36	85.33	21.20	89.56	88.01	97.84	11.57	75.30	87.28	98.49	6.98	73.20	99.52	99.63	0.24	92.48	78.09	82.12	65.98	69.10
+ TIDE	94.85	83.96	18.86	89.56	91.13	98.72	5.52	74.60	93.15	99.00	3.80	74.70	99.61	99.70	0.10	91.98	78.61	82.90	44.97	68.70

6.6. Extended OOD Exposure Study

With OOD exposure regularisation (Eq. 4), TIDE achieves superior or competitive performance. Shown in Table 6, TIDE outperforms GNNSAFE++ and NODESAFE++ across multiple datasets, as highlighted in bold. For example, on Cora, TIDE achieves an AUROC of 95.76 (vs. 93.13 & 91.32) and significantly reduces the FPR95 from 42.48 down to 19.34. On larger Pubmed, TIDE improves AUROC to 98.26 (vs. 93.77 & 95.14) and decreases FPR95 to 8.84 (vs. 39.58 & 24.87). On more challenging real-world OOD datasets, TIDE remains competitive on the Twitch and Arxiv datasets. This highlights TIDE’s adaptability to incorporate OOD exposure strategies, enabling it to achieve superior performance.

6.7. Comparison with Advanced Graph OOD Detectors

To further demonstrate the effectiveness of TIDE and the role of the IB objective, we compare against two advanced graph OOD detection baselines, DeGEM (Chen et al., 2025) and GOLD (Wang et al., 2025a), on Cora and Pubmed under structure shift, feature shift, and label shift settings in Table 5. All results were reproduced to the best of our ability. Both baselines exhibit strong performance across several settings, reflecting the strength of their respective designs. To enable a fair comparison with our decomposition-based method, we additionally evaluate “+TIDE” variants of GOLD and DeGEM by incorporating our TIDE objective during training. These TIDE-augmented versions often improve detection over their base models, demonstrating the benefit and efficacy of our information-decomposition framework. Additional comparison with disentanglement-based methods are shown in Appendix 6.5.

7. Conclusion

We propose TIDE, a novel tri-component information decomposition framework for OOD detection on graph-structured data. By decomposing information into structural, feature, and joint components with an IB objective, TIDE preserves shift-indicative joint-input label information while suppressing label-irrelevant and spurious correlations in individual input components. We provide theoretical insights showing that the IB principle improves in-distribution con-

Table 6. Comparisons of OOD exposure with **best** and **runner-up** highlighted. GS(NS)++ short for GNNSafe(NODESAFE)++.

	Metrics	OOD Exposure				TIDE w/ OE
		OE	Energy FT	GS++	NS++	
Cora	AUROC (↑)	79.76	85.13	93.13	91.32	95.76
	AUPR (↑)	64.93	67.89	85.28	82.74	89.68
	FPR95 (↓)	75.22	51.03	37.28	42.48	19.34
	ID ACC (↑)	77.69	80.44	82.16	74.94	82.20
Citeseer	AUROC (↑)	63.75	79.81	85.69	84.29	92.16
	AUPR (↑)	47.20	52.79	65.89	64.86	75.90
	FPR95 (↓)	74.15	57.37	54.76	56.12	29.22
	ID ACC (↑)	62.24	72.66	72.74	68.60	75.81
Pubmed	AUROC (↑)	78.38	76.25	93.77	95.14	98.26
	AUPR (↑)	27.67	27.61	74.06	72.20	85.36
	FPR95 (↓)	79.05	91.02	39.58	24.87	8.84
	ID ACC (↑)	73.00	75.80	77.88	74.17	76.67
Twitch	AUROC (↑)	55.72	84.50	95.76	76.79	91.52
	AUPR (↑)	70.18	88.04	97.45	83.97	93.59
	FPR95 (↓)	95.07	61.29	29.81	34.46	33.19
	ID ACC (↑)	70.73	70.52	70.36	70.07	68.74
Arxiv	AUROC (↑)	69.80	71.56	73.98	74.50	74.89
	AUPR (↑)	80.15	80.47	82.50	81.55	83.25
	FPR95 (↓)	85.16	80.59	79.99	77.53	77.37
	ID ACC (↑)	52.39	53.26	53.28	51.32	52.41

fidence compared to standard supervised learning, which in turn benefits OOD detection. Extensive experiments validate the efficacy of TIDE, which outperforms SOTA SL-based OOD detection methods, including both non-OOD-exposed and OOD-exposed approaches.

Impact Statement

We hope our work inspires future research on node-level graph OOD detection in real-world settings. As a foundational study on OOD detection for graph-structured data, we do not identify any direct negative societal impacts. Our research relies on publicly available datasets, algorithms, and models, all of which are properly acknowledged and pose no risks requiring safeguards. While potential societal implications may exist, none warrant specific emphasis in this study.

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Appendix

A. Reproducibility Statement

To support reproducible research, we summarise our efforts as below:

1. **Baselines & Datasets.** We follow the baseline from (Wu et al., 2023b) and utilise publicly available datasets. The details are described in Section 6 and Appendix L.
2. **Model training.** Detailed implementation setting is provided in Section 6 and Appendix K.
3. **Evaluation Metrics.** We discuss the evaluation metrics used in Section 6 and Appendix J.
4. **Methodology.** Our TIDE framework is fully documented in Section 4, with implementation notes provided in Section 4.3. To support implementation and reproducibility we provide a detailed discussion on obtaining the **tractable objective** in Appendix F, a **neural network parameterisation** of TIDE is given in Appendix G. In addition, we provide a detailed pseudo code of the optimisation process in Algorithm 1 and the inference process in Algorithm 2. We provide theoretical analysis and corresponding proofs in Section 5 and Appendix C.

B. Potential Limitations

In Section 4.1, we illustrate that TIDE employs two additional networks on top of the GNN classifier to effectively capture structural and feature information: a GNN for modelling structural relationships and an MLP for processing feature-based information. This multi-network architecture inevitably incurs additional computational and memory overhead during training. However, during inference, our model achieves comparable inference speeds while delivering significant performance improvements across all evaluated datasets. We argue that the increased training cost is a justifiable trade-off given the substantial gains in detection performance. The detailed computational cost is discussed in Appendix N. Described in Appendix K, a constant α and β values were applied uniformly in the current experiments, future work will explore fine-grained control of individual terms. Furthermore, currently implementation only considers a simple structure representation learning strategy (i.e., using a constant feature vector), we will investigate the use of more advanced position encoding and structure representation learning techniques in future work. This may include methods like spectral embeddings via Laplacian eigenmaps, random walk-based encodings, and learnable position encodings etc. Since our TIDE focuses on exploring the advantages of IB over standard supervised learning, we have left the comparison and integration with more advanced graph OOD detection methods and novel training frameworks like DeGEM and GOLD as future work (Chen et al., 2025; Wang et al., 2025a). Moreover, our proposed method and theoretical analysis are currently focused on node-level classification and we do not study the anomaly detection task. However, we believe the framework can be extended to graph-level OOD detection. Potential directions include applying the IB principle to the representation of entire graphs or learning compact and informative subgraphs, as explored in prior works (Dai et al., 2023; Sun et al., 2022). We also aim to investigate the impact of this framework in the context of heterophilic graphs. These extensions present promising avenues for future research.

C. Proof of Technical Insights

C.1. Proof of Proposition 4.1

Proof. Given $\mathbf{Z}$ is encoded from $(\mathbf{X}, \mathbf{A})$ by the network f_θ . Hence, $\mathbf{Z}$ depends only on $(\mathbf{X}, \mathbf{A})$ or $\mathbf{Z} \perp \mathbf{Y} \mid (\mathbf{X}, \mathbf{A})$, and the Markov chain $\mathbf{Y} \rightarrow (\mathbf{X}, \mathbf{A}) \rightarrow \mathbf{Z}$ holds. Equivalently, this implies $p(z \mid x, a, y) = p(z \mid x, a)$ and we have $I(\mathbf{Z}; \mathbf{Y} \mid \mathbf{X}, \mathbf{A}) = 0$. More explicitly,

$$I(\mathbf{Z}; \mathbf{Y} \mid \mathbf{X}, \mathbf{A}) = \mathbb{E}_{p(x,a)} [D_{\text{KL}}(p(z, y \mid x, a) \parallel p(z \mid x, a)p(y \mid x, a))].$$

Under the above Markov chain, $p(z, y \mid x, a) = p(z \mid x, a)p(y \mid x, a)$, so the KL term is zero, which gives $I(\mathbf{Z}; \mathbf{Y} \mid \mathbf{X}, \mathbf{A}) = 0$. Then, by the chain rule of mutual information,

$$I(\mathbf{X}, \mathbf{A}, \mathbf{Z}; \mathbf{Y}) = I(\mathbf{X}, \mathbf{A}; \mathbf{Y}) + I(\mathbf{Z}; \mathbf{Y} \mid \mathbf{X}, \mathbf{A}) = I(\mathbf{X}, \mathbf{A}; \mathbf{Y}).$$

□

C.2. Proof of Lemma 5.1

Proof. Consider a deterministic encoder with encoded representation $\mathbf{Z} = f(\mathbf{X}, \mathbf{A})$, the conditional mutual information $I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y})$ can be expanded as:

$$\begin{aligned} I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y}) &= H(\mathbf{X}, \mathbf{A} | \mathbf{Y}) + H(\mathbf{Z} | \mathbf{Y}) - H(\mathbf{X}, \mathbf{A}, \mathbf{Z} | \mathbf{Y}) \\ &= H(\mathbf{X}, \mathbf{A} | \mathbf{Y}) + H(\mathbf{Z} | \mathbf{Y}) - (H(\mathbf{X}, \mathbf{A} | \mathbf{Y}) + H(\mathbf{Z} | \mathbf{X}, \mathbf{A}, \mathbf{Y})) \\ &= H(\mathbf{Z} | \mathbf{Y}) - H(\mathbf{Z} | \mathbf{X}, \mathbf{A}, \mathbf{Y}). \end{aligned} \quad (15)$$

Since $\mathbf{Z}$ is determined by $\mathbf{X}$ and $\mathbf{A}$, the second term vanishes:

$$H(\mathbf{Z} | \mathbf{X}, \mathbf{A}, \mathbf{Y}) = 0 \quad \Rightarrow \quad I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y}) = H(\mathbf{Z} | \mathbf{Y}). \quad (16)$$

Thus, minimising $I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y})$ is equivalent to minimising $H(\mathbf{Z} | \mathbf{Y})$.

Notably, reducing $H(\mathbf{Z} | \mathbf{Y})$ implies the conditional distribution $P(\mathbf{Z} | \mathbf{Y})$ becomes more concentrated (i.e., $\mathbf{Z}$ is more predictable given $\mathbf{Y}$ (reduced uncertainty)). Applying Bayes' theorem, we have:

$$P(\mathbf{Y} | \mathbf{Z}) = \frac{P(\mathbf{Z} | \mathbf{Y})P(\mathbf{Y})}{P(\mathbf{Z})}. \quad (17)$$

Here, a more concentrated $P(\mathbf{Z} | \mathbf{Y})$ would amplify the likelihood term $P(\mathbf{Z} | \mathbf{Y})$ relative to the marginal $P(\mathbf{Z}) = \mathbb{E}_{\mathbf{Y}'}[P(\mathbf{Z} | \mathbf{Y}')]$. As a result, it produces sharper peaks in $P(\mathbf{Y} | \mathbf{Z})$, increasing the maximum probability $\max_{\mathbf{y}} P(\mathbf{Y} = \mathbf{y} | \mathbf{Z})$ and thus the prediction confidence of ID data, where:

$$H(\mathbf{Y} | \mathbf{Z}) = -\mathbb{E}_{\mathbf{Z}} \left[\sum_{\mathbf{y}} P(\mathbf{y} | \mathbf{Z}) \log P(\mathbf{y} | \mathbf{Z}) \right] \rightarrow 0. \quad (18)$$

□

C.3. Proof of Theorem 5.2

Proof. To begin, using the chain rule of mutual information, we can decompose the second mutual information term in the IB objective as:

$$I(\mathbf{X}, \mathbf{A}; \mathbf{Z}) = I(\mathbf{Z}; \mathbf{Y}) + I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y}). \quad (19)$$

Substituting into the IB objective yields:

$$\max(1 - \beta)I(\mathbf{Z}; \mathbf{Y}) - \beta I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y}). \quad (20)$$

Notably, when $\beta = 0$, the objective reduces to the standard supervised learning goal of maximising $I(\mathbf{Z}; \mathbf{Y})$. However, this indicates that the superfluous information $I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y})$ is preserved, meaning the encoded representation $\mathbf{Z}$ may contain information from $\mathbf{X}$ and $\mathbf{A}$ that is irrelevant to the label $\mathbf{Y}$. This can negatively impact predictive performance, as $\mathbf{Z}$ is not optimised to focus solely on label-relevant information.

In contrast, for $\beta \in (0, 1)$, the IB objective simultaneously increases predictive information $I(\mathbf{Z}; \mathbf{Y})$ and reduces superfluous information by minimising $I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y})$. This ensures $\mathbf{Z}$ can capture useful information about $\mathbf{Y}$, while compressing out input information irrelevant to $\mathbf{Y}$. From Lemma 5.1, minimising $I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y})$ reduces $H(\mathbf{Z} | \mathbf{Y})$. This reduction forces $\mathbf{Z}$ to discard information in $\mathbf{X}$ that is irrelevant to $\mathbf{Y}$, leading to a sharper conditional distribution $P(\mathbf{Y} | \mathbf{Z})$. Thus, when $P(\mathbf{Y} | \mathbf{Z})$ becomes sharper, $\max_{\mathbf{y}} P(\mathbf{Y} = \mathbf{y})$ increases, as the probability mass is concentrated on the most likely value of $\mathbf{Y}$. Consequently, the conditional entropy decreases:

$$H(\mathbf{Y} | \mathbf{Z}) = -\mathbb{E}_{\mathbf{Z}} [\max_{\mathbf{y}} P(\mathbf{Y} = \mathbf{y} | \mathbf{Z})] + \text{cross-entropy terms}, \quad (21)$$

thereby increasing prediction confidence. In other words, the model becomes more certain about its predictions, as $\mathbf{Z}$ is optimised to focus on the most relevant information for predicting $\mathbf{Y}$.

Additionally, an optimal trade-off occurs when β balances information compression-to-prediction. Let $\Delta_Z = \text{Var}(I(\mathbf{Z}; \mathbf{Y}))$ and $\Delta_{Z|Y} = \text{Var}(I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y}))$ represent parameter sensitivity. The critical ratio:

$$\beta < \frac{\Delta_Z}{\Delta_Z + \Delta_{Z|Y}} \quad (22)$$

ensures sufficient weight on predictive information $I(\mathbf{Z}; \mathbf{Y})$. Under this condition, IB produces representations $\mathbf{Z}$ with minimised superfluous information and maximised prediction confidence than standard supervised learning. $\square$

Derivation of $I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y}) = I(\mathbf{X}; \mathbf{Z} | \mathbf{Y}) + I(\mathbf{A}; \mathbf{Z} | \mathbf{Y}, \mathbf{X})$. The decomposition follows from the chain rule of mutual information:

$$\begin{aligned} I(\mathbf{X}, \mathbf{A}; \mathbf{Z} | \mathbf{Y}) &= I(\mathbf{X}; \mathbf{Z} | \mathbf{Y}) + I(\mathbf{A}; \mathbf{Z} | \mathbf{Y}, \mathbf{X}) \\ &= H(\mathbf{Z} | \mathbf{Y}) - H(\mathbf{Z} | \mathbf{Y}, \mathbf{X}) \\ &\quad + H(\mathbf{Z} | \mathbf{Y}, \mathbf{X}) - H(\mathbf{Z} | \mathbf{Y}, \mathbf{X}, \mathbf{A}) \\ &= H(\mathbf{Z} | \mathbf{Y}) - H(\mathbf{Z} | \mathbf{Y}, \mathbf{X}, \mathbf{A}). \end{aligned} \quad (23)$$

For deterministic encoders where $\mathbf{Z} = f(\mathbf{X}, \mathbf{A})$, we have $H(\mathbf{Z} | \mathbf{X}, \mathbf{A}, \mathbf{Y}) = 0$, by Eq. 16, we complete the derivation.

C.4. Proof of Proposition 5.3

Proof. Given an optimised network f^* trained with ID data $(\mathbf{X}_{\text{id}}, \mathbf{A}_{\text{id}})$ via the IB objective. Let $\mathbf{Z}^* = f^*(\mathbf{X}, \mathbf{A})$ be the encoded representation obtained from the network. We can derive the following:

Part 1: ID Data Compression From the derivation in Eq. 23, given ID inputs $(\mathbf{X}_{\text{id}}, \mathbf{A}_{\text{id}})$ and optimal encoded ID representation $\mathbf{Z}_{\text{id}}^*$, the second term in the IB objective in Eq. 36 can be expressed as:

$$I(\mathbf{X}_{\text{id}}, \mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{id}}^* | \mathbf{Y}) \quad (24)$$

Following directly from Lemma 5.1 and Theorem 5.2, this is equivalent to minimising $H(\mathbf{Z}_{\text{id}}^* | \mathbf{Y})$, making $P(\mathbf{Y} | \mathbf{Z}_{\text{id}}^*)$ sharply concentrated. Thus:

$$H(\mathbf{Y} | \mathbf{Z}_{\text{id}}^*) = -\mathbb{E}_{\mathbf{Z}_{\text{id}}^*} \left[\sum_y P(y | \mathbf{Z}_{\text{id}}^*) \log P(y | \mathbf{Z}_{\text{id}}^*) \right] \rightarrow 0. \quad (25)$$

Part 2: OOD Data Separation To investigate the effect on OOD data, without loss of generality, we consider the feature shift defined in Section I, where $P_{\text{id}}(\mathbf{X}) \neq P_{\text{ood}}(\mathbf{X})$, $P_{\text{id}}(\mathbf{X}|\mathbf{A}) \neq P_{\text{ood}}(\mathbf{X}|\mathbf{A})$, $P_{\text{id}}(\mathbf{Y}|\mathbf{X}) \neq P_{\text{ood}}(\mathbf{Y}|\mathbf{X})$, and $P_{\text{id}}(\mathbf{Y}|\mathbf{A}, \mathbf{X}) \neq P_{\text{ood}}(\mathbf{Y}|\mathbf{A}, \mathbf{X})$. Such that we have OOD features $\mathbf{X}_{\text{ood}}$ with ID structure $\mathbf{A}_{\text{id}}$ (i.e., $\mathcal{G}_{\text{feat shift}} = (\mathbf{X}_{\text{ood}}, \mathbf{A}_{\text{id}})$). Let $\mathbf{Z}_{\text{ood}}^* = f^*(\mathbf{X}_{\text{ood}}, \mathbf{A}_{\text{id}})$ denote the representation encoded from the ID trained model for the given OOD data.

From the derivation in Eq. 23, we have:

$$I(\mathbf{X}_{\text{id}}, \mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{id}}^* | \mathbf{Y}) = I(\mathbf{X}_{\text{id}}; \mathbf{Z}_{\text{id}}^* | \mathbf{Y}) + I(\mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{id}}^* | \mathbf{Y}, \mathbf{X}_{\text{id}}). \quad (26)$$

$$I(\mathbf{X}_{\text{ood}}, \mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{ood}}^* | \mathbf{Y}) = I(\mathbf{X}_{\text{ood}}; \mathbf{Z}_{\text{ood}}^* | \mathbf{Y}) + I(\mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{ood}}^* | \mathbf{Y}, \mathbf{X}_{\text{ood}}). \quad (27)$$

Suppose for contradiction that:

$$I(\mathbf{X}_{\text{ood}}; \mathbf{Z}_{\text{ood}}^* | \mathbf{Y}) < I(\mathbf{X}_{\text{id}}; \mathbf{Z}_{\text{id}}^* | \mathbf{Y}). \quad (28)$$

This would imply:

$$I(\mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{ood}}^* | \mathbf{Y}, \mathbf{X}_{\text{ood}}) \approx I(\mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{id}}^* | \mathbf{Y}, \mathbf{X}_{\text{id}}), \quad (29)$$

$$\Rightarrow I(\mathbf{X}_{\text{ood}}, \mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{ood}}^* | \mathbf{Y}) \leq I(\mathbf{X}_{\text{id}}, \mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{id}}^* | \mathbf{Y}). \quad (30)$$

However, this contradicts the IB optimality condition since the encoder $\mathbf{Z}^*$ was never trained to compress $\mathbf{X}_{\text{ood}}$. Thus, we must have:

$$I(\mathbf{X}_{\text{ood}}; \mathbf{Z}_{\text{ood}}^* | \mathbf{Y}) \geq I(\mathbf{X}_{\text{id}}; \mathbf{Z}_{\text{id}}^* | \mathbf{Y}). \quad (31)$$

Similarly, since $\mathbf{A}_{\text{id}}$ is fixed but $\mathbf{X}_{\text{ood}}$ is novel:

$$I(\mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{ood}}^* \mid \mathbf{Y}, \mathbf{X}_{\text{ood}}) \geq I(\mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{id}}^* \mid \mathbf{Y}, \mathbf{X}_{\text{id}}). \quad (32)$$

Therefore:

$$I(\mathbf{X}_{\text{ood}}, \mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{ood}}^* \mid \mathbf{Y}) \geq I(\mathbf{X}_{\text{id}}, \mathbf{A}_{\text{id}}; \mathbf{Z}_{\text{id}}^* \mid \mathbf{Y}). \quad (33)$$

From Lemma 5.1, higher conditional mutual information implies higher $H(\mathbf{Z}_{\text{ood}}^* \mid \mathbf{Y})$. By Bayes' theorem:

$$P(\mathbf{Y} \mid \mathbf{Z}_{\text{ood}}^*) = \frac{P(\mathbf{Z}_{\text{ood}}^* \mid \mathbf{Y})P(\mathbf{Y})}{P(\mathbf{Z}_{\text{ood}}^*)}. \quad (34)$$

The diffuse $P(\mathbf{Z}_{\text{ood}}^* \mid \mathbf{Y})$ makes $P(\mathbf{Y} \mid \mathbf{Z}_{\text{ood}}^*)$ approximately uniform, yielding:

$$H(\mathbf{Y} \mid \mathbf{Z}_{\text{ood}}^*) \approx \log C \gg H(\mathbf{Y} \mid \mathbf{Z}_{\text{id}}^*), \quad (35)$$

where C is the number of classes. This entropy gap enables improved OOD detection through logit-based scoring methods (i.e., energy score (Liu et al., 2020), MaxLogit (Hendrycks et al., 2022)). $\square$

D. Graph Neural Network

Graph Neural Networks (GNNs) are inherently well-suited for capturing intricate dependencies between nodes in a graph. Their effectiveness largely stems from the message-passing mechanism, which progressively aggregates information from neighbouring nodes, allowing the model to learn both local and global structural and feature patterns. Let $\mathbf{z}_i^{(l)}$ represent the learned embedding of node i at layer l . A standard Graph Convolutional Network (GCN) updates node representations iteratively using the propagation rule:

$$\mathbf{Z}^{(l)} = \sigma \left(\mathbf{D}^{-1/2} \tilde{\mathbf{A}} \mathbf{D}^{-1/2} \mathbf{Z}^{(l-1)} \mathbf{W}^{(l)} \right),$$

where $\mathbf{Z}^{(l-1)} = [\mathbf{z}_i^{(l-1)}]$, and the initial node features are given by $\mathbf{H}^{(0)} = \mathbf{X}$. Here, $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with self-loops, $\mathbf{I}$ denotes the identity matrix, $\mathbf{D}$ is the diagonal degree matrix of $\tilde{\mathbf{A}}$, σ is a nonlinear activation function (e.g., ReLU), and $\mathbf{W}^{(l)}$ represents the trainable weight matrix for layer l (Kipf & Welling, 2017).

E. Information Bottleneck Principle

The information bottleneck (IB) principle aims to learn a compressed representation $\mathbf{Z}$ with the maximum relevant information about the label $\mathbf{Y}$ (i.e., $I(\mathbf{Z}; \mathbf{Y})$) while minimising the label-irrelevant information (i.e., $I(\mathbf{X}; \mathbf{Z})$), as constrained by the Markov chain $\mathbf{Y} \rightarrow \mathbf{X} \rightarrow \mathbf{Z}$ (Tishby et al., 2000; Alemi et al., 2017). For graph data, $\mathbf{Z}$ is a function of both node features $\mathbf{X}$ and graph structure $\mathbf{A}$ (i.e., $\mathbf{Z} = \text{GNN}(\mathbf{X}, \mathbf{A})$). The IB objective can be defined as:

$$\max_{\mathbf{Z}} I(\mathbf{Z}; \mathbf{Y}) - \beta I(\mathbf{X}, \mathbf{A}; \mathbf{Z}), \quad (36)$$

where β is a Lagrange multiplier controlling the trade-off between compression and prediction. This enables the compression of the joint input information to representation $\mathbf{Z}$ via $I(\mathbf{X}, \mathbf{A}; \mathbf{Z})$ and optimise its classification ability via $I(\mathbf{Z}; \mathbf{Y})$ (Wu et al., 2020; Sun et al., 2022).

F. Tractable Optimisation of TIDE

To optimise the intractable TIDE objective in Eq. 13, we approximate the IB terms via variational lower bounds (Alemi et al., 2017) and enforce the conditional independence and pairwise mutual information minimisation using reconstruction loss and contrastive loss (Cheng et al., 2020), respectively. **Variational Approximation of $\mathcal{L}_{\text{IB}}$:** Beginning with the first term in objective Eq. 13, without loss of generality, we provide a variational approximation bound for $\text{IB}_{\mathbf{Z}} = I(\mathbf{Z}; \mathbf{Y}) - \beta_{\mathbf{Z}} I(\mathbf{X}, \mathbf{A}; \mathbf{Z})$, and the tractable bounds for $\text{IB}_{\mathbf{V}}$ and $\text{IB}_{\mathbf{Q}}$ can be derived accordingly. Let $q(\mathbf{Y}|\mathbf{Z})$ and $r(\mathbf{Z})$

denote variational approximations to the true conditional distribution $p_{\mathbf{Z}}(\mathbf{Y}|\mathbf{Z})$, and marginal distribution $p(\mathbf{Z})$, respectively. Consider a parametric Gaussian distribution as prior $p(\mathbf{Z}|\mathbf{X}, \mathbf{A})$, we have:

$$p(\mathbf{Z}|\mathbf{X}, \mathbf{A}) = \mathcal{N}(\mathbf{Z}; \mu(f(\mathbf{X}, \mathbf{A})), \Sigma(f(\mathbf{X}, \mathbf{A}))),$$

where f is modelled as a GNN network that encodes the input $(\mathbf{X}, \mathbf{A})$, followed by linear layers to obtain the mean and variance representations respectively. Subsequently, we apply the reparameterisation trick as $\mathbf{Z} = f(\mathbf{X}, \mathbf{A}, \epsilon)$, which ensures it is a deterministic function of $\mathbf{X}, \mathbf{A}$ and the Gaussian random variable $\epsilon \sim p(\epsilon) = \mathcal{N}(\mathbf{0}, \mathbf{1})$. Thus, given training data $\mathbf{X}_{\text{tr}} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$, $\mathbf{Y}_{\text{tr}} = (y_1, \dots, y_N)$ with adjacency matrix $\mathbf{A}_{\text{tr}}$, using the empirical data distribution, we can obtain a tractable variational lower bound for $\text{IB}_{\mathbf{Z}}$:

$$\text{VIB}_{\mathbf{Z}} = \frac{1}{N} \sum_{n=1}^N [\mathbb{E}_{\epsilon \sim p(\epsilon)} [\log q(y_n | f(\mathbf{x}_n, \mathbf{A}, \epsilon))] - \beta \text{KL}(p(\mathbf{Z}|\mathbf{x}_n, \mathbf{A}) || r(\mathbf{Z}))] \leq \text{IB}_{\mathbf{Z}}, \quad (37)$$

where $r(\mathbf{Z}) = \mathcal{N}(\mathbf{Z}, \mathbf{0}, \mathbf{1})$ is fixed to the standard normal. The first term in Eq. 37 represents the log-likelihood of the output $\mathbf{Y}$ given the representation $\mathbf{Z}$, which can be calculated using the cross entropy loss to encourage $\mathbf{Z}$ to be predictive of $\mathbf{Y}$. The second term is the Kullback-Leibler (KL) divergence between the conditional distribution $p(\mathbf{Z} | \mathbf{X}, \mathbf{A})$ and the variational prior $r(\mathbf{Z})$, which reduces the irrelevant information compressed from the joint input $(\mathbf{X}, \mathbf{A})$ into the representation $\mathbf{Z}$. Following a similar approach, we can derive the lower bound for $\text{IB}_{\mathbf{V}}$ and $\text{IB}_{\mathbf{Q}}$, and thus obtain the tractable variational lower bound $\mathcal{L}_{\text{VIB}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}}$ for the first term in Eq. 13 as:

$$\max \mathcal{L}_{\text{IB}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}} \geq \mathcal{L}_{\text{VIB}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}} = \text{VIB}_{\mathbf{Z}} + \text{VIB}_{\mathbf{V}} + \text{VIB}_{\mathbf{Q}}. \quad (38)$$

Conditional Independence of $\mathcal{L}_{\text{CInd}}$: To make $\mathcal{L}_{\text{CInd}} = I(\mathbf{A}; \mathbf{X}|\mathbf{Z})$ tractable, we can use a variational approximation $q(\mathbf{X}|\mathbf{Z})$ to estimate the true distribution $p(\mathbf{X}|\mathbf{Z})$ and derive a variational upper bound via:

$$\min \mathcal{L}_{\text{VCInd}_{\mathbf{X}, \mathbf{A}, \mathbf{Z}}} = \mathbb{E}_{p(\mathbf{x}, \mathbf{A}, \mathbf{z})} [\underbrace{\log p(\mathbf{x}|\mathbf{A}, \mathbf{z})}_{(1)} - \underbrace{\log q(\mathbf{x}|\mathbf{z})}_{(2)}] \geq \mathcal{L}_{\text{CInd}_{\mathbf{X}, \mathbf{A}, \mathbf{Z}}}. \quad (39)$$

Minimising this objective encourages $\mathbf{Z}$ to encode all the information of $\mathbf{X}$ that is relevant to $\mathbf{A}$ (via (1)) and reducing the information of $\mathbf{X}$ that is independent of $\mathbf{A}$ (via (2)). This ensures $\mathbf{X}$ and $\mathbf{A}$ is conditionally independent given $\mathbf{Z}$.

Pairwise Mutual Information Minimisation via Contrastive Learning $\mathcal{L}_{\text{PMI}}$: Similar to the variational bound for the IB terms, without loss of generality, we provide an upper bound for $I(\mathbf{Z}; \mathbf{V})$, and the bounds for $I(\mathbf{Z}; \mathbf{Q})$ and $I(\mathbf{V}; \mathbf{Q})$ can be derived accordingly. Notably, since the conditional distribution $p(\mathbf{Z}|\mathbf{V})$ is unknown, utilising the variational contrastive log-ratio upper bound of mutual information (CLUB) (Cheng et al., 2020) with samples $(\mathbf{z}_n, \mathbf{v}_n)$, we can derive a tractable upper bound for $I(\mathbf{Z}; \mathbf{V})$:

$$I_{\text{VCLUB}_{\mathbf{Z}, \mathbf{V}}} = \frac{1}{N} \sum_{n=1}^N [\log q(\mathbf{z}_n | \mathbf{v}_n) - \frac{1}{N} \sum_{m=1}^N \log q(\mathbf{z}_m | \mathbf{v}_n)] \geq I(\mathbf{Z}; \mathbf{V}), \quad (40)$$

for $q(\mathbf{Z}|\mathbf{V})$ is a variational distribution to approximate $p(\mathbf{Z}|\mathbf{V})$. The first term evaluates how well the variational approximation predicts the “positive” pairs $(\mathbf{z}_n, \mathbf{v}_n)$, capturing the dependence between $\mathbf{Z}$ and $\mathbf{V}$. The second term penalises this by evaluating the model on “negative” pairs $(\mathbf{z}_m, \mathbf{v}_n)$, approximating the behaviour when $\mathbf{Z}$ and $\mathbf{V}$ are independent, thereby providing an effective upper bound for $I(\mathbf{Z}; \mathbf{V})$. Similarly, we can derive the upper bounds for $I(\mathbf{Z}; \mathbf{Q})$ and $I(\mathbf{V}; \mathbf{Q})$, and obtain the final tractable objective $\mathcal{L}_{\text{VPMI}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}}$ as:

$$\min \mathcal{L}_{\text{VPMI}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}} = \alpha_1 I_{\text{VCLUB}_{\mathbf{Z}, \mathbf{V}}} + \alpha_2 I_{\text{VCLUB}_{\mathbf{Z}, \mathbf{Q}}} + \alpha_3 I_{\text{VCLUB}_{\mathbf{V}, \mathbf{Q}}} \geq \mathcal{L}_{\text{PMI}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}}. \quad (41)$$

Hence, minimising this loss will ensure the pairwise MI independence between $\mathbf{Z}$, $\mathbf{V}$, and $\mathbf{Q}$. **Final tractable TIDE objective:** Combining Eq. 38, 39, 41, the overall tractable objective for TIDE is given by:

$$\max_{\theta_{\mathbf{Z}}, \theta_{\mathbf{V}}, \theta_{\mathbf{Q}}} \mathcal{L}_{\text{VIB}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}} - \lambda_{\text{CInd}} \mathcal{L}_{\text{VCInd}_{\mathbf{X}, \mathbf{A}, \mathbf{Z}}} - \mathcal{L}_{\text{VPMI}_{\mathbf{Z}, \mathbf{V}, \mathbf{Q}}}. \quad (42)$$

While the objective in Eq. 42 combines all losses into one expression, we optimise the networks individually using their respective losses.

The optimisation process for TIDE is outlined in Algorithm 1 in the main text. In **Step 1**, the individual networks are updated to capture sufficient yet minimal joint-input, structure-only, and feature-only information for predicting the label $\mathbf{Y}$. **Step 2** enforces mutual conditional independence between $\mathbf{X}|\mathbf{Z}$ and $\mathbf{A}$ specifically for the GNN classifier. Finally, **Step 3** minimises pairwise mutual information, updating each network only with the loss relevant to its representation. This structured approach ensures efficient and targeted optimisation for TIDE.

G. Neural Network Parameterisation of TIDE

To optimise the intractable TIDE objective in Eq. 42, we parameterise the variational approximations using GNNs and MLPs: **Main Encoder Networks.** Our framework uses three primary networks to encode the representations $\mathbf{Z}$, $\mathbf{V}$ and $\mathbf{Q}$ and for prediction (i.e., joint-input classifier, feature network, structure network) with parameters $\theta_{\mathbf{Z}}$, $\theta_{\mathbf{V}}$, and $\theta_{\mathbf{Q}}$ respectively:

$$\text{Joint-input Network: } \text{GNN}_{\text{CLS}}(\mathbf{X}, \mathbf{A}) \leftarrow \mathbf{Z} \quad (43)$$

$$\text{Feature Network: } \text{MLP}_{\text{feat}}(\mathbf{X}) \leftarrow \mathbf{V} \quad (44)$$

$$\text{Structure Network: } \text{GNN}_{\text{struct}}(\mathbf{I}, \mathbf{A}) \leftarrow \mathbf{Q}, \quad (45)$$

where $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the node feature matrix, $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the adjacency matrix, $\mathbf{I} \in \mathbb{R}^{n \times n}$ is the identity matrix (used as placeholder features), and $\mathbf{Z}$, $\mathbf{V}$, $\mathbf{Q}$ are the latent representations. To capture structural information, we leverage a GNN model that inherently provides robust structure encoding. Meanwhile, for the feature network - comprising solely feature embeddings - a lightweight MLP is more appropriate.

Each network contains the following elements respectively: **Variational Implementation.** Following common variational approximations (Alemi et al., 2017), we model the conditional distributions of our latent representations using Gaussian distributions parameterised by the respective encoders. We use a GNN/MLP encoder to extract representations $\mathbf{h}$ from the given inputs, and use linear MLP layers to encode the latent variables μ and σ , this produces Gaussian distributions over the latent space:

$$\mu_{\mathbf{Z}} = \text{MLP}_{\mu_{\mathbf{Z}}}(\mathbf{h}_{\mathbf{Z}}), \quad \sigma_{\mathbf{Z}} = \text{MLP}_{\sigma_{\mathbf{Z}}}(\mathbf{h}_{\mathbf{Z}}), \quad \mathbf{h}_{\mathbf{Z}} = \text{GNN}_{\text{enc}}(\mathbf{X}, \mathbf{A}) \quad (46)$$

$$\mu_{\mathbf{V}} = \text{MLP}_{\mu_{\mathbf{V}}}(\mathbf{h}_{\mathbf{V}}), \quad \sigma_{\mathbf{V}} = \text{MLP}_{\sigma_{\mathbf{V}}}(\mathbf{h}_{\mathbf{V}}), \quad \mathbf{h}_{\mathbf{V}} = \text{MLP}_{\text{enc}}(\mathbf{X}) \quad (47)$$

$$\mu_{\mathbf{Q}} = \text{MLP}_{\mu_{\mathbf{Q}}}(\mathbf{h}_{\mathbf{Q}}), \quad \sigma_{\mathbf{Q}} = \text{MLP}_{\sigma_{\mathbf{Q}}}(\mathbf{h}_{\mathbf{Q}}), \quad \mathbf{h}_{\mathbf{Q}} = \text{GNN}_{\text{enc}}(\mathbf{I}, \mathbf{A}). \quad (48)$$

Using these, we can sample the latent variables using the reparameterisation trick:

$$\mathbf{Z} = \mu_{\mathbf{Z}} + \sigma_{\mathbf{Z}} \odot \epsilon_{\mathbf{Z}}, \quad \epsilon_{\mathbf{Z}} \sim \mathcal{N}(0, \mathbf{I}) \quad (49)$$

$$\mathbf{V} = \mu_{\mathbf{V}} + \sigma_{\mathbf{V}} \odot \epsilon_{\mathbf{V}}, \quad \epsilon_{\mathbf{V}} \sim \mathcal{N}(0, \mathbf{I}) \quad (50)$$

$$\mathbf{Q} = \mu_{\mathbf{Q}} + \sigma_{\mathbf{Q}} \odot \epsilon_{\mathbf{Q}}, \quad \epsilon_{\mathbf{Q}} \sim \mathcal{N}(0, \mathbf{I}) \quad (51)$$

This enable us to obtain the representations $\mathbf{Z}$, $\mathbf{V}$ and $\mathbf{Q}$ for optimisation.

Auxiliary Layers. For predictions and calculation of the reconstruction loss, IB minimisation, and pairwise MI minimisation, we use the following auxiliary layers in the respective networks:

$$\text{Prediction Layers: } \text{GNN}_{\text{pred-Z}}(\mathbf{Z}, \mathbf{A}) \rightarrow \hat{\mathbf{Y}}_{\mathbf{Z}} \quad (52)$$

$$\text{MLP}_{\text{pred-V}}(\mathbf{V}) \rightarrow \hat{\mathbf{Y}}_{\mathbf{V}} \quad (53)$$

$$\text{GNN}_{\text{pred-Q}}(\mathbf{Q}, \mathbf{A}) \rightarrow \hat{\mathbf{Y}}_{\mathbf{Q}} \quad (54)$$

$$\text{Reconstruction Layer: } \text{MLP}_{\text{recon}}(\mathbf{Z}) \rightarrow \hat{\mathbf{X}} \quad (55)$$

G.1. Objective Components in Terms of Networks

1) Information Bottleneck Terms. For each representation $\mathbf{Z}$, $\mathbf{V}$, and $\mathbf{Q}$, we calculate the VIB loss Eq. 37. For example, for the joint representation $\mathbf{Z}$:

$$\begin{aligned} \text{VIB}_{\mathbf{Z}} = & \underbrace{\mathbb{E}_{\epsilon_{\mathbf{Z}}} [\log \text{MLP}_{\text{pred-}\mathbf{Z}}(\mathbf{Z} = \boldsymbol{\mu}_{\mathbf{Z}} + \boldsymbol{\sigma}_{\mathbf{Z}} \odot \epsilon_{\mathbf{Z}})(\mathbf{Y})]}_{\text{CE}(\mathbf{Z}, \mathbf{Y})} \\ & - \underbrace{\beta_{\mathbf{Z}} \sum_i (1 + \log((\boldsymbol{\sigma}_{\mathbf{Z},i})^2) - (\boldsymbol{\mu}_{\mathbf{Z},i})^2 - (\boldsymbol{\sigma}_{\mathbf{Z},i})^2)}_{\text{KL divergence term (analytical form)}} \end{aligned} \quad (56)$$

2) Conditional Independence Term. For the conditional independence loss Eq. 39, we use the reconstruction network:

$$\mathcal{L}_{\text{CInd}} = \underbrace{\mathbb{E}_{\epsilon_{\mathbf{Z}}} [-\|\mathbf{X} - \text{MLP}_{\text{recon}}(\mathbf{Z} = \boldsymbol{\mu}_{\mathbf{Z}} + \boldsymbol{\sigma}_{\mathbf{Z}} \odot \epsilon_{\mathbf{Z}})\|_2^2]}_{\text{Reconstruction loss (encourages } \mathbf{Z} \text{ to encode } \mathbf{X} \text{ information)}} \quad (57)$$

3) Pairwise Mutual Information Terms. Following CLUB (Cheng et al., 2020), we estimate the pairwise mutual information terms using contrastive learning Eq. 41. For each pair of representations (i.e., $\mathbf{Z}$, $\mathbf{V}$), we compute:

$$\text{CLUB}(\mathbf{Z}, \mathbf{V}) = \frac{1}{N} \sum_{i=1}^N \log \frac{s(\mathbf{Z}_i, \mathbf{V}_i)}{\frac{1}{N} \sum_{j=1}^N s(\mathbf{Z}_i, \mathbf{V}_j)} \quad (58)$$

$$(59)$$

where s is a similarity function (e.g., dot product) between representations, N is the number of samples.

Implementation Notes In practice, we optimise this objective by:

- Forward pass through all encoders to get $\boldsymbol{\mu}_{\mathbf{Z}}, \boldsymbol{\sigma}_{\mathbf{Z}}, \boldsymbol{\mu}_{\mathbf{V}}, \boldsymbol{\sigma}_{\mathbf{V}}, \boldsymbol{\mu}_{\mathbf{Q}}, \boldsymbol{\sigma}_{\mathbf{Q}}$.
- Sample $\mathbf{Z}$, $\mathbf{V}$, $\mathbf{Q}$ using the reparameterisation trick.
- Compute all components of the loss using these samples.
- Backpropagate through the networks to update parameters according to Algorithm 1.

H. Extended Related Work

Out-of-distribution detection is a critical task in machine learning, extensively studied across various domains. A significant body of work focuses on methods that rely solely on ID data, employing techniques such as softmax scores (Hendrycks & Gimpel, 2017; Liang et al., 2018), energy-based scoring (Liu et al., 2020; Wang et al., 2021; Yang et al., 2024b), and activation pruning (Djurisic et al., 2023; Sun & Li, 2022; Sun et al., 2021). Other strategies enhance model confidence (Hsu et al., 2020; Hein et al., 2019; Vyas et al., 2018), improve feature learning (Lin et al., 2021; Dong et al., 2022), or incorporate adversarial approaches (Bitterwolf et al., 2020; Chen et al., 2021; Choi & Chung, 2020). Beyond ID-based methods, OOD exposure leverages auxiliary OOD data during training to improve detection performance (Hendrycks et al., 2019; Liu et al., 2020; Park et al., 2023; Zhu et al., 2023; Zheng et al., 2023; Du et al., 2024; Wu et al., 2023b). Meanwhile, an emerging direction involves generating synthetic OOD data. GAN-based methods, such as ConfOOD (Lee et al., 2018a), train confidence classifiers alongside OOD data generation, while VOS (Du et al., 2022) synthesises outliers from low-probability Gaussian regions. More recently, diffusion models have been widely adopted, as seen in DFDD (Wu et al., 2023a) and Dream-OOD (Du et al., 2023).

H.1. Graph OOD Detection

Recent advancements in OOD detection have expanded to graph-structured data, addressing both node-level and graph-level detection tasks. For node-level OOD detection, several innovative methods have been developed to improve detection accuracy and robustness. GNNSafe introduces an energy propagation schema that considers the inter-dependence of node

instances, providing a more nuanced approach to identifying OOD nodes (Wu et al., 2023b). Building on this, NODESafe incorporates additional regularization terms to reduce and bound extreme energy scores, ensuring more stable and reliable detection (Yang et al., 2024b). Meanwhile, TopoOOD explores topological shifts in graph data and proposes a node-wise Dirichlet Energy metric to measure neighbourhood turbulence, which serves as a confidence score for OOD detection (Bao et al., 2024). Other approaches, such as GKDE, employ a multi-source uncertainty framework to estimate node-level Dirichlet distributions, which aids in identifying OOD instances (Zhao et al., 2020). Similarly, GPN leverages Bayesian posterior and density estimation to quantify uncertainty at the node level, further enhancing detection capabilities (Stadler et al., 2021). Moreover, DeGEM presents a novel energy-based model training framework involving a multi-hop graph encoder and energy head, targeting the detection on heterophilic graphs (Chen et al., 2025). Additionally, GOLD presents a data-synthesis-based framework that generates pseudo-OOD embeddings without relying on pre-trained generative models or auxiliary OOD datasets. At its core is an alternating optimisation framework, which effectively balances ID representation learning with divergence-enhanced pseudo-OOD generation (Wang et al., 2025a). RAGNOR presents a post-hoc baseline that leverages embedding norms for OOD detection on both node-level and graph-level tasks (Gu et al., 2025). More recently, some works explore OOD detection on text-attributed graph using Large Language Models for zero-shot detection or generating auxiliary pseudo-OOD samples (Xu et al., 2025b;a; Wang et al., 2024).

At the graph level, OOD detection methods have focused on modelling distribution shifts, adopting data-centric perspectives, and utilising unsupervised learning techniques. GraphDE models distribution shifts through a graph generative process, deriving a posterior distribution to detect OOD graphs (Li et al., 2022b). In contrast, AAGOD takes a data-centric approach by learning structural patterns in graph data through a learnable amplifier matrix, improving detection performance (Guo et al., 2023). Another notable method, GOOD-D, applies unsupervised contrastive learning to enhance graph-level OOD detection without relying on labelled OOD data (Liu et al., 2023). These methods highlight the growing emphasis on leveraging graph structure and distributional properties to improve OOD detection.

H.2. Information Bottleneck

The IB principle has been widely applied across various domains to enhance learned representations (Wu et al., 2020; Alemi et al., 2017; Tishby et al., 2000; Ahuja et al., 2021; Kawaguchi et al., 2023; Wang et al., 2023; Federici et al., 2020; Dai et al., 2023; Sun et al., 2022). Its objective is to learn a compressed representation that maximally retains label-relevant information while minimising label-irrelevant information from inputs (Tishby et al., 2000; Alemi et al., 2017). This concept has also been extended to graph-structured data, enabling robust representation learning from both node features and graph structure (Wu et al., 2020; Sun et al., 2022).

Notably, IB’s potential for OOD detection has primarily been explored in Euclidean settings (Hu et al., 2024; Zhao & Cao, 2023; Li et al., 2023b; Sinha et al., 2021; Wu & Deng, 2024). For instance, DRL introduces a dual representation learning framework that learns both a target-discriminative representation and an additional distribution-discriminative representation C , capturing all information relevant to the target Y (Zhao & Cao, 2023). Meanwhile, (Hu et al., 2024) examines information from the perspective of classification-relevant and classification-irrelevant detection, providing a theoretical analysis of the overconfidence of OOD samples in models trained on ID data using supervised learning. (Alemi et al., 2018) empirically validates IB’s effectiveness for OOD detection. In contrast to these studies, our work offers theoretical insights into the advantages of IB over standard supervised learning, particularly in improving prediction confidence. For graph-structured data, IS-GIB introduces I-GIB to mitigate irrelevant information by minimising the mutual information between the input graph and its embeddings. S-GIB was further leveraged to utilise structural relationships to discard irrelevant information, establishing an effective invariant learning framework for OOD generalisation (Yang et al., 2024a). Additionally, CSIB generates and refines causal subgraphs based on invariant causal prediction and the graph information bottleneck principle, preserving essential features while filtering out spurious correlations, thereby improving graph-based OOD generalisation (An et al., 2024). Moreover, IBPL introduces a graph-level OOD detection method that effectively tackles the issue of overlapping features between ID and OOD graphs, aiming to enhance detection performance. IBPL proposes a novel graph prompt that jointly optimises node features and graph structure, enabling the generation of more discriminative ID features. By leveraging the IB principle, IBPL maximises the MI between category labels and the prompt graph while minimising the MI between perturbed graphs and the prompt graph. This dual optimisation process allows for the extraction of robust ID features while significantly reducing the influence of overlapping features (Cao et al., 2025). Unlike these approaches, our work is driven by theoretical insights into IB and mutual decomposition, demonstrating their benefits for OOD detection. We validate our findings through extensive experiments, highlighting and validating the effectiveness of IB for graph OOD detection.

I. Defining OOD shifts

Typically, considering a message passing neural network (i.e., GNN) model, it is trained using ID data (i.e., $\mathbf{X}_{\text{ID}}^{tr}, \mathbf{A}_{\text{ID}}^{tr}$), and the test data consists of a combination of ID and OOD instances (i.e., $(\mathbf{X}_{\text{ID}}^{te}, \mathbf{A}_{\text{ID}}^{te}), (\mathbf{X}_{\text{OOD}}^{te}, \mathbf{A}_{\text{OOD}}^{te})$). The primary objective is to identify and detect the OOD samples from the ID instances while maintaining high ID classification performance.

Generally, we categorise OOD shifts into the following: $P_{\text{train}}(\mathbf{X}, \mathbf{A}) \neq P_{\text{test}}(\mathbf{X}, \mathbf{A})$ and the conditional distribution $P_{\text{train}}(\mathbf{Y}|\mathbf{X}, \mathbf{A}) \neq P_{\text{test}}(\mathbf{Y}|\mathbf{X}, \mathbf{A})$. To detect OOD data, the task is to formulate an OOD scoring function F , usually built upon the output from the classifier GNN, such that it outputs $F(\mathbf{X}, \mathbf{A}; \text{GNN}) = 0$ for data from in-distribution and $F(\mathbf{X}, \mathbf{A}; \text{GNN}) = 1$ for data from out-of-distribution. This definition, however, lacks the detail of the particular shift occurring alone on the feature $\mathbf{X}$ or among the structure $\mathbf{A}$. Thus, to understand when OOD detection is possible for such an ID-trained model, we adopt the following definitions and explore the impact of each of the distribution shifts (Han et al., 2024). **Feature Shift ($\mathbf{X}$):** A feature shift occurs when the distribution of $\mathbf{X}$ and its relationship with $\mathbf{A}$ change between training and testing. Specifically, $P_{\text{train}}(\mathbf{X}) \neq P_{\text{test}}(\mathbf{X})$ and $P_{\text{train}}(\mathbf{X}|\mathbf{A}) \neq P_{\text{test}}(\mathbf{X}|\mathbf{A})$ or $P_{\text{train}}(\mathbf{A}|\mathbf{X}) \neq P_{\text{test}}(\mathbf{A}|\mathbf{X})$. Additionally, the conditional distribution of $\mathbf{Y}$ given $\mathbf{X}$ changes, such that $P_{\text{train}}(\mathbf{Y}|\mathbf{X}) \neq P_{\text{test}}(\mathbf{Y}|\mathbf{X})$. Consequently, the joint conditional distribution of $\mathbf{Y}$ given both $\mathbf{X}$ and $\mathbf{A}$ also shifts: $P_{\text{train}}(\mathbf{Y}|\mathbf{A}, \mathbf{X}) \neq P_{\text{test}}(\mathbf{Y}|\mathbf{A}, \mathbf{X})$. However, the conditional distribution of $\mathbf{Y}$ given $\mathbf{A}$ alone remains unchanged: $P_{\text{train}}(\mathbf{Y}|\mathbf{A}) = P_{\text{test}}(\mathbf{Y}|\mathbf{A})$. **Structural Shift ($\mathbf{A}$):** A structural shift occurs when the distribution of $\mathbf{A}$ and dependencies with $\mathbf{X}$ shifts (i.e., $P_{\text{train}}(\mathbf{A}) \neq P_{\text{test}}(\mathbf{A})$ and $P_{\text{train}}(\mathbf{X}|\mathbf{A}) \neq P_{\text{test}}(\mathbf{X}|\mathbf{A})$ or $P_{\text{train}}(\mathbf{A}|\mathbf{X}) \neq P_{\text{test}}(\mathbf{A}|\mathbf{X})$). Additionally, the conditional distribution of $\mathbf{Y}$ given $\mathbf{A}$ changes, such that $P_{\text{train}}(\mathbf{Y}|\mathbf{A}) \neq P_{\text{test}}(\mathbf{Y}|\mathbf{A})$. Consequently, the joint conditional distribution of $\mathbf{Y}$ given both $\mathbf{X}$ and $\mathbf{A}$ also shifts: $P_{\text{train}}(\mathbf{Y}|\mathbf{A}, \mathbf{X}) \neq P_{\text{test}}(\mathbf{Y}|\mathbf{A}, \mathbf{X})$. However, the conditional distribution of $\mathbf{Y}$ given $\mathbf{X}$ remains invariant: $P_{\text{train}}(\mathbf{Y}|\mathbf{X}) = P_{\text{test}}(\mathbf{Y}|\mathbf{X})$. **Joint Shift ($\mathbf{X}, \mathbf{A}$):** More realistically, distributional shifts typically occur jointly on $\mathbf{X}, \mathbf{A}$, thus, we consider a joint distribution shift on both the feature and structure (i.e., $P_{\text{train}}(\mathbf{A}) \neq P_{\text{test}}(\mathbf{A})$, $P_{\text{train}}(\mathbf{X}) \neq P_{\text{test}}(\mathbf{X})$, and $P_{\text{train}}(\mathbf{X}|\mathbf{A}) \neq P_{\text{test}}(\mathbf{X}|\mathbf{A})$ or $P_{\text{train}}(\mathbf{A}|\mathbf{X}) \neq P_{\text{test}}(\mathbf{A}|\mathbf{X})$). The joint conditional distribution of $\mathbf{Y}$ given both $\mathbf{X}$ and $\mathbf{A}$ also shifts: $P_{\text{train}}(\mathbf{Y}|\mathbf{A}, \mathbf{X}) \neq P_{\text{test}}(\mathbf{Y}|\mathbf{A}, \mathbf{X})$. **Semantic Shift ($\mathbf{Y}$):** Consider $\mathcal{Y}_{\text{train}}$ and $\mathcal{Y}_{\text{test}}$ as the label space of the train and test data respectively. OOD data with semantic shift is defined to consist of unknown labels $\mathbf{Y}$ that do not belong to any of the classes seen in the ID label space $\mathcal{Y}_{\text{train}}$, i.e., $\mathcal{Y}_{\text{train}} \subset \mathcal{Y}_{\text{test}}$ or $\mathcal{Y}_{\text{train}} \cap \mathcal{Y}_{\text{test}} = \emptyset$.

J. Evaluation Metrics

To evaluate the performance of OOD detection, we followed common practices (Wu et al., 2023b; Liu et al., 2020; 2023; Yang et al., 2024b) and utilised three key metrics:

1. Area Under the Receiver Operating Characteristic curve (AUROC),
2. Area Under the Precision-Recall curve (AUPR),
3. False positive rate at 95% true positive rate (FPR95).

These metrics are independent of the threshold, avoiding the need to select τ . AUROC captures the balance between the true positive rate (TPR) and false positive rate (FPR) across varying thresholds, offering an overall assessment of the model’s ability to differentiate between ID and OOD samples. However, when OOD instances are rare in highly imbalanced datasets, AUROC can yield overly optimistic results. In contrast, AUPR accounts for both precision and recall, making it more suited for such imbalanced scenarios. Meanwhile, FPR95 emphasises performance under high-sensitivity conditions by measuring the rate at which ID samples are misclassified as OOD when the true positive rate is fixed at 95%. This metric highlights improvements in detection performance under stricter criteria, where a lower FPR95 indicates a more significant gain in the model’s ability to differentiate between ID and OOD instances.

K. Implementation Details

We follow (Wu et al., 2023b; Yang et al., 2024b) and use the publicly available benchmark datasets. The datasets were downloaded via Pytorch Geometric 2.0.3 and OGB 1.3.3 under the MIT license. Experiments were conducted using Python 3.9.19 and PyTorch 2.3.1 with Cuda 12.2 on a single NVIDIA RTX A6000 GPU with 48GB of memory. We follow (Wu et al., 2023b; Yang et al., 2024b) for the selection of baselines. We conducted experiments across three seeds for GNNSAFE and NODESAFE. Result inconsistencies with original reported statistics may stem from software environment, dataset

differences, and unavailable hyperparameters. We reproduced results to the best of our ability, while results for other baselines were sourced from (Wu et al., 2023b; Yang et al., 2024b).

Our design choices for the auxiliary networks directly follow their intended roles in TIDE’s decomposition framework: **V** should capture feature-only information, and **Q** should capture structure-only information. The architectures are chosen to enforce this disentanglement rather than model preference.

- **Feature network V = MLP (captures feature-only information)**

The purpose of **V** is to isolate feature-specific components that do not rely on graph propagation. We feed only the raw node attributes into an MLP to ensure that depends solely on features and does not inadvertently encode structural context. A GNN with identity adjacency could simulate “no propagation,” but this would introduce unnecessary complexity. A simple MLP is therefore the appropriate choice.

- **Structure network Q = GCN (captures structure-only information)**

The purpose of **Q** is to encode predictive information arising purely from graph topology. To enforce this, we use a GCN with constant node features (all-ones input). This ensures that **Q** encodes only structural variation from the adjacency matrix. More expressive alternatives (learnable positional encodings, spectral embeddings, structure-only encoders) can be explored in future work.

CLUB (Cheng et al., 2020) is used for implementing the PMI regulariser. However, we also compare the effect with alternative MI estimator like InfoNCE (van den Oord et al., 2018) in Table 10, where the performance remains consistent. This indicates that TIDE’s performance is driven by the overall information decomposition design rather than any specific MI estimator, and that optimisation remains stable under reasonable perturbations of these approximations. The trade-off parameters β for the individual IB losses are searched through the range $\beta \in \{0.0001, 0.001, 0.01, 0.1, 1\}$, with a default value of $\beta = 0.001$. A constant $\alpha_i \in \{0.0001, 0.001, 0.01, 0.1, 1\}$ is applied uniformly to all three mutual information terms in Eq. 11. The loss coefficient λ_{CInd} for $\mathcal{L}_{\text{VCInd}}$ was tuned based on the dataset with values taking ranges of $\lambda_{\text{CInd}} \in \{0.0001, \dots, 1\}$. Following (Wu et al., 2023b), the number of energy propagation iterations k is set to 2, with the controlling parameter α fixed at 0.5, the OOD-exposure loss coefficient λ_{OE} for $\mathcal{L}_{\text{EReg}}$ is set to 1. For OOD exposure experiments, we tuned the margins t_{ID} and t_{OOD} from the proposed ranges by (Wu et al., 2023b; Yang et al., 2024b), if not available, we tune with values from the range of $\{-9, \dots, 0\}$ for $t_{\text{ID}} < t_{\text{OOD}}$ for different datasets. The Adam optimiser is used for training (Kingma & Ba, 2015). For simplicity, constant β and α values were applied uniformly: α to all mutual information terms in PMI loss, and β to IB terms for the GNN classifier, structure, and feature networks. Results show performance is sensitive to these weights. A suitable β is crucial for balancing representation robustness with prediction ability, while an appropriate α encourages a compact joint representation, mitigating spurious impacts from structure and features. Tables 7, 8, and 9 presents the hyperparameter sensitivity analysis.

Table 7. Hyperparameter analysis for λ_{CInd} . Bold highlights the optimal parameter.

λ_{CInd}	Cora - S				Citeseer - F			
	AUROC	AUPR	FPR	ID Acc	AUROC	AUPR	FPR	ID Acc
0	94.79 $\pm$ 0.22	88.41 $\pm$ 0.49	26.22 $\pm$ 2.55	78.73	93.25 $\pm$ 0.87	82.94 $\pm$ 2.15	26.96 $\pm$ 4.64	67.83
0.0001	95.01 $\pm$ 0.24	88.93 $\pm$ 0.51	24.84 $\pm$ 0.70	78.90	93.96 $\pm$ 0.73	84.67 $\pm$ 2.35	23.13 $\pm$ 3.55	68.40
0.001	94.97 $\pm$ 0.19	88.89 $\pm$ 0.44	25.55 $\pm$ 1.86	78.80	93.81 $\pm$ 0.81	84.51 $\pm$ 2.42	24.58 $\pm$ 3.31	68.37
0.01	94.93 $\pm$ 0.32	88.80 $\pm$ 0.69	25.39 $\pm$ 0.70	78.90	93.79 $\pm$ 0.83	84.53 $\pm$ 2.44	24.75 $\pm$ 4.58	68.33
0.1	94.92 $\pm$ 0.26	88.77 $\pm$ 0.60	25.65 $\pm$ 1.80	78.90	93.74 $\pm$ 0.78	84.25 $\pm$ 2.67	24.99 $\pm$ 4.20	68.37
1	95.15 $\pm$ 0.37	89.33 $\pm$ 0.57	23.31 $\pm$ 1.04	78.97	93.90 $\pm$ 0.75	84.62 $\pm$ 2.32	24.05 $\pm$ 3.94	68.30

Table 8. Hyperparameter analysis for α – weight coefficients applied uniformly to all mutual information terms in PMI. Bold highlights the optimal parameter.

α	Cora - F				PubMed - S			
	AUROC	AUPR	FPR	ID Acc	AUROC	AUPR	FPR	ID Acc
0	97.72 $\pm$ 0.10	94.26 $\pm$ 0.52	9.40 $\pm$ 1.06	77.47	96.45 $\pm$ 0.64	76.46 $\pm$ 2.31	15.66 $\pm$ 4.88	74.77
0.0001	97.68 $\pm$ 0.13	89.16 $\pm$ 9.49	8.80 $\pm$ 0.79	77.77	96.55 $\pm$ 0.80	77.26 $\pm$ 2.72	16.21 $\pm$ 5.18	74.47
0.001	97.66 $\pm$ 0.10	94.51 $\pm$ 0.40	8.63 $\pm$ 0.94	78.00	96.54 $\pm$ 0.80	77.21 $\pm$ 2.71	16.15 $\pm$ 4.98	74.47
0.01	97.88 $\pm$ 0.24	94.67 $\pm$ 0.56	8.13 $\pm$ 1.32	78.10	97.25 $\pm$ 0.58	79.76 $\pm$ 2.17	12.97 $\pm$ 3.87	75.93
0.1	78.49 $\pm$ 7.47	70.99 $\pm$ 8.49	91.89 $\pm$ 6.77	76.30	88.13 $\pm$ 6.57	66.19 $\pm$ 11.57	69.39 $\pm$ 30.40	75.70
1	47.82 $\pm$ 4.92	29.19 $\pm$ 3.29	97.89 $\pm$ 1.31	24.37	37.96 $\pm$ 49.44	31.37 $\pm$ 49.70	78.33 $\pm$ 33.53	40.77

Table 9. Hyperparameter analysis for β – prediction-to-compression trade-off weight applied uniformly to the IB terms for the GNN classifier, structure, and feature networks. Bold highlights the optimal parameter.

β	Amazon - L				Twitch			
	AUROC	AUPR	FPR	ID Acc	AUROC	AUPR	FPR	ID Acc
0	96.92 $\pm$ 0.33	96.38 $\pm$ 0.70	9.29 $\pm$ 1.15	96.10	66.33 $\pm$ 15.32	72.59 $\pm$ 13.44	81.18 $\pm$ 19.87	70.75
0.0001	96.85 $\pm$ 0.13	95.82 $\pm$ 0.11	8.94 $\pm$ 1.45	95.89	89.72 $\pm$ 5.42	91.92 $\pm$ 3.85	46.60 $\pm$ 26.47	68.63
0.001	96.83 $\pm$ 0.39	95.78 $\pm$ 0.42	8.36 $\pm$ 3.83	96.04	89.39 $\pm$ 5.57	91.58 $\pm$ 3.82	47.75 $\pm$ 27.79	68.63
0.01	96.86 $\pm$ 0.35	95.79 $\pm$ 0.39	8.13 $\pm$ 3.55	96.07	89.04 $\pm$ 6.61	91.61 $\pm$ 4.75	48.89 $\pm$ 33.03	68.61
0.1	96.87 $\pm$ 0.31	95.80 $\pm$ 0.38	8.39 $\pm$ 3.17	94.87	82.14 $\pm$ 13.18	85.87 $\pm$ 9.48	51.07 $\pm$ 33.40	67.52
1	93.40 $\pm$ 1.77	90.85 $\pm$ 1.20	34.01 $\pm$ 30.72	82.32	57.70 $\pm$ 24.03	68.52 $\pm$ 14.31	68.30 $\pm$ 25.88	59.31

Table 10. Comparison of different mutual information estimators. Best is highlighted in bold.

Dataset	Method	AUROC	AUPR	FPR	ACC
Pubmed-S	w/ CLUB	97.25	79.76	12.97	75.93
	w/ InfoNCE	97.33	77.61	11.69	74.30
Cora-S	w/ CLUB	95.15	89.33	23.31	78.97
	w/ InfoNCE	94.95	88.86	24.78	78.20
Cora-F	w/ CLUB	97.88	94.67	8.13	78.10
	w/ InfoNCE	97.94	94.38	6.94	78.10
Citeseer-L	w/ CLUB	90.97	64.92	30.84	88.55
	w/ InfoNCE	90.69	63.28	31.34	87.54
Coauthor-S	w/ CLUB	99.95	99.94	0.13	92.72
	w/ InfoNCE	99.94	99.92	0.13	92.66

L. Dataset Details

Following the protocol outlined in (Wu et al., 2023b), we use publicly available graph benchmark datasets, sourced from the PyTorch Geometric (PyG) and the Open Graph Benchmark (OGB) ¹ package² (Sen et al., 2008). We adhere to the provided splits and dataset generation process described in (Wu et al., 2023b). **Cora** This dataset represents a citation network where nodes correspond to academic papers, and edges denote citation relationships (Sen et al., 2008). Each paper is classified into one of seven categories. Cora lacks explicit domain-based partitions for OOD evaluation, thus, we follow (Wu et al., 2023b) and generate OOD data synthetically as described in Section 6 (i.e., Structure shift, Feature shift, and label-leave-out).

Table 11. Cora dataset statistics

	Structure (ID)	Structure (OOD)	Feature (ID)	Feature (OOD)	Label (ID)	Label (OOD)
Nodes	2708	2708	2708	2708	904	986
Edges	10556	6696	10556	10556	10556	10556
Feature Dim	1433	1433	1433	1433	1433	1433
Classes	7	7	7	7	3	3

Citeseer This dataset is another citation network (Giles et al., 1998; Sen et al., 2008), where nodes represent scientific papers classified into one of six classes, and edges denote citation relationships. It contains slightly more papers with fewer edges than Cora, however, the feature dimension is larger. We follow Cora and generate three OOD data synthetically as described in Section 6.

Pubmed This dataset is a biomedical paper citation network (Sen et al., 2008), with each paper classified into one of three classes. The nodes represent academic papers, while edges are citation relationships, we follow (Yang et al., 2024b) and generate OOD data synthetically as described in Section 6. Due to the Pubmed dataset having only three classes, the

¹<https://github.com/snap-stanford/ogb?tab=readme-ov-file>

²<https://pytorch-geometric.readthedocs.io/en/latest/modules/datasets.html>

Table 12. Citeseer dataset statistics

	Structure (ID)	Structure (OOD)	Feature (ID)	Feature (OOD)	Label (ID)	Label (OOD)
Nodes	3327	3327	3327	3327	1104	1522
Edges	9104	5932	9104	9104	9104	9104
Feature Dim	3703	3703	3703	3703	3703	3703
Classes	6	6	6	6	2	3

NODESAFE approach designates two classes as OOD labels (i.e., OOD training and OOD testing), leaving just one class in the training set. This setup may pose challenges, as training with only one class can lead to significant imbalance and limitations in model evaluation. As a result, we exclude the label shift scenario for Pubmed from our analysis.

Table 13. Pubmed dataset statistics

	Structure (ID)	Structure (OOD)	Feature (ID)	Feature (OOD)
Nodes	19717	19717	19717	19717
Edges	88648	74188	88648	88648
Feature Dim	500	500	500	500
Classes	3	3	3	3

Amazon-Photo This dataset models an item co-purchasing network, where nodes represent products, and edges indicate frequently co-purchased items (McAuley et al., 2015). Node features capture product descriptions, and labels correspond to product categories. Similar to Cora, we generate OOD data synthetically. **Coauthor-CS** This dataset represents

Table 14. Amazon-Photo dataset statistics

	Structure (ID)	Structure (OOD)	Feature (ID)	Feature (OOD)	Label (ID)	Label (OOD)
Nodes	7650	7650	7650	7650	3095	3673
Edges	238162	149168	238162	238162	238162	238162
Feature Dim	745	745	745	745	745	745
Classes	8	8	8	8	3	4

a collaboration network, where nodes correspond to authors, and edges indicate co-authorships in computer science research (Sinha et al., 2015). The task involves classifying authors into their respective fields based on publication keywords. OOD graphs are generated following the synthetic protocol used for other datasets.

Table 15. Coauthor-CS dataset statistics

	Structure (ID)	Structure (OOD)	Feature (ID)	Feature (OOD)	Label (ID)	Label (OOD)
Nodes	18333	18333	18333	18333	13290	3649
Edges	163788	92802	163788	163788	163788	163788
Feature Dim	6805	6805	6805	6805	6805	6805
Classes	15	15	15	15	10	4

TwitchGamers - Explicit This dataset comprises multiple social network subgraphs from different geographic regions (Rozemberczki & Sarkar, 2021). Nodes represent Twitch gamers, and edges indicate follower relationships. Node features include game-based embeddings, and the classification task focuses on predicting whether a user streams mature content. We use the DE subgraph as ID data and the ES, FR, and RU subgraphs as OOD test data. **OGBN-Arxiv** This dataset is a large-scale citation network spanning research papers from 1960 to 2020 (Hu et al., 2020). Nodes represent papers, categorised by subject area, and edges signify citation links. Node features are derived from word embeddings of paper titles and abstracts. Following (Wu et al., 2023b), we partition the dataset using publication timestamps—papers published before 2015 are used as ID data, while papers published after 2017 serve as OOD data.

Table 16. TwitchGamers - Explicit dataset statistics

	DE (ID)	ES (OOD)	FR (OOD)	RU (OOD)
Nodes	9498	4648	6551	4385
Edges	315774	123412	231883	78993
Feature Dim	128	128	128	128
Classes	2	2	2	2

Table 17. OGBN-Arxiv dataset statistics

	2015 (ID)	2018 (OOD)	2019 (OOD)	2020 (OOD)
Nodes	53160	29799	39711	8892
Edges	152226	622466	1061197	1166243
Feature Dim	128	128	128	128
Classes	40	40	40	40

M. Extended Experimental Results

In this section, we provide the extended results from the main paper. Table 18. Additionally, we provide the OOD detection performance for each subset of the OOD datasets in Tables 19 to 25, complementing Tables 1 and 6 in the main text. The scores reported in the subset tables are averaged across three runs, with variance reflecting performance deviation across the three seeds. Furthermore, we report an extended version of the ablation study in Tables 26 to 30, supplementing Table 3 in the main text.

Table 18. Overall Model performance comparison: out-of-distribution detection is measured by **AUROC** ($\uparrow$) / **AUPR** ($\uparrow$) / **FPR95** ($\downarrow$) (%) and in-distribution classification results are measured by accuracy (**ID ACC**) ($\uparrow$). OOD detection performance was prioritised, with the detection results of our TIDE against Non- (Real-) OOD Exposure methods.

	Metrics	Non-OOD Exposure								Real OOD Exposure				TIDE	
		MSP	ODIN	Maha	Energy	GKDE	GPN	GNNSAFE	NODESAFE	OE	Energy FT	GNNSAFE++	NODESAFE++	w/o OE	w/ OE
Cora	AUROC	82.55	49.87	54.74	83.09	69.54	84.56	91.20 $\pm$ 3.09	93.35 $\pm$ 2.41	79.76	85.13	93.13 $\pm$ 2.62	91.32 $\pm$ 1.55	95.58 $\pm$ 2.12	95.76 $\pm$ 2.29
	AUPR	65.82	26.08	34.43	66.21	46.09	68.02	82.92 $\pm$ 4.96	84.64 $\pm$ 6.40	64.93	67.89	85.28 $\pm$ 4.55	82.74 $\pm$ 4.62	89.34 $\pm$ 5.32	89.68 $\pm$ 5.95
	FPR95	62.39	100.00	96.30	65.21	80.51	58.30	50.53 $\pm$ 22.04	29.23 $\pm$ 12.06	75.22	51.03	37.28 $\pm$ 16.97	42.48 $\pm$ 5.82	20.09 $\pm$ 10.72	19.34 $\pm$ 11.00
	ID ACC	79.91	79.61	79.57	80.34	79.86	81.65	81.56 $\pm$ 7.01	82.66 $\pm$ 6.61	77.69	80.44	82.16 $\pm$ 7.78	74.94 $\pm$ 13.86	82.56 $\pm$ 6.99	82.2 $\pm$ 6.79
Citeseer	AUROC	77.69	50.14	49.55	78.26	72.95	78.22	84.89 $\pm$ 5.47	87.80 $\pm$ 2.74	63.75	79.81	85.69 $\pm$ 5.16	84.29 $\pm$ 7.06	92.27 $\pm$ 1.53	92.16 $\pm$ 1.53
	AUPR	51.43	21.38	29.29	51.22	47.67	52.22	64.86 $\pm$ 3.39	72.35 $\pm$ 6.41	47.20	52.79	65.89 $\pm$ 2.50	64.86 $\pm$ 4.66	76.57 $\pm$ 10.34	75.90 $\pm$ 10.27
	FPR95	69.42	100	95.06	65.34	71.85	67.09	60.85 $\pm$ 18.06	53.38 $\pm$ 17.47	74.15	57.37	54.76 $\pm$ 23.14	56.12 $\pm$ 22.56	30.79 $\pm$ 7.64	29.22 $\pm$ 7.10
	ID ACC	73.72	73.75	62.17	73.43	72.69	72.77	76.06 $\pm$ 12.05	73.12 $\pm$ 14.15	62.24	72.66	72.74 $\pm$ 13.43	68.60 $\pm$ 18.51	75.66 $\pm$ 11.19	75.81 $\pm$ 10.95
Pubmed	AUROC	78.80	49.72	62.20	79.25	78.14	78.76	93.82 $\pm$ 1.93	91.08 $\pm$ 2.95	78.38	76.25	93.77 $\pm$ 1.31	95.14 $\pm$ 0.26	97.35 $\pm$ 0.13	98.26 $\pm$ 0.29
	AUPR	28.37	4.83	11.74	28.21	24.65	28.65	69.94 $\pm$ 6.32	68.56 $\pm$ 1.06	27.67	27.61	74.06 $\pm$ 5.16	72.20 $\pm$ 5.85	80.73 $\pm$ 1.37	85.36 $\pm$ 0.21
	FPR95	76.73	100	91.26	70.69	75.04	71.06	37.71 $\pm$ 14.28	50.17 $\pm$ 0.89	79.05	91.02	39.58 $\pm$ 11.44	24.87 $\pm$ 6.44	12.75 $\pm$ 0.31	8.84 $\pm$ 1.81
	ID ACC	75.05	75.30	71.15	75.55	74.65	75.15	76.78 $\pm$ 0.31	77.32 $\pm$ 0.02	73.00	75.80	77.88 $\pm$ 0.35	74.17 $\pm$ 0.05	76.17 $\pm$ 0.33	76.67 $\pm$ 0.19
Amazon	AUROC	96.52	80.12	73.81	96.73	66.98	92.60	97.99 $\pm$ 0.93	97.84 $\pm$ 1.11	97.79	98.04	—	—	98.16 $\pm$ 1.14	—
	AUPR	95.01	77.18	72.35	95.16	71.18	90.50	98.16 $\pm$ 1.56	97.77 $\pm$ 2.11	97.26	96.96	—	—	98.08 $\pm$ 1.99	—
	FPR95	13.83	85.22	83.44	13.15	98.47	32.64	3.24 $\pm$ 5.24	3.69 $\pm$ 6.14	7.52	5.98	—	—	2.81 $\pm$ 4.61	—
	ID ACC	93.83	93.88	93.80	93.85	87.71	89.54	93.79 $\pm$ 1.99	92.70 $\pm$ 2.16	93.54	93.38	—	—	93.90 $\pm$ 1.88	—
Coauthor	AUROC	95.74	51.71	82.02	96.64	69.24	69.89	98.98 $\pm$ 1.41	99.02 $\pm$ 1.45	97.65	98.17	—	—	99.27 $\pm$ 1.17	—
	AUPR	96.43	56.37	87.05	97.09	80.17	72.77	99.55 $\pm$ 0.44	99.57 $\pm$ 0.47	98.04	98.51	—	—	99.70 $\pm$ 0.40	—
	FPR95	21.37	99.97	48.09	15.49	97.04	69.60	4.29 $\pm$ 6.87	4.33 $\pm$ 7.04	10.61	7.76	—	—	3.31 $\pm$ 5.45	—
	ID ACC	93.37	93.29	93.29	93.57	87.74	89.39	93.65 $\pm$ 1.50	93.21 $\pm$ 1.86	93.41	93.44	—	—	93.48 $\pm$ 1.74	—
Twitch	AUROC	33.59	58.16	55.68	51.24	46.48	51.73	66.33 $\pm$ 15.32	66.48 $\pm$ 15.44	55.72	84.50	95.76 $\pm$ 2.63	76.79 $\pm$ 34.23	89.72 $\pm$ 5.42	91.52 $\pm$ 6.53
	AUPR	49.14	72.12	66.42	60.81	62.11	66.36	72.59 $\pm$ 13.44	72.71 $\pm$ 13.43	70.18	88.04	97.45 $\pm$ 1.69	83.97 $\pm$ 24.11	91.92 $\pm$ 3.85	93.59 $\pm$ 4.36
	FPR95	97.45	93.96	90.13	91.61	95.62	95.51	81.18 $\pm$ 19.87	80.62 $\pm$ 20.03	95.07	61.29	29.81 $\pm$ 23.16	34.46 $\pm$ 29.93	46.60 $\pm$ 26.47	33.19 $\pm$ 29.86
	ID ACC	68.72	70.79	70.51	70.40	67.44	68.09	70.75	70.75	70.73	70.52	70.36	70.07	68.63	68.74
Arxiv	AUROC	63.91	55.07	56.92	64.20	58.32	OOM	70.58 $\pm$ 6.41	71.36 $\pm$ 6.35	69.80	71.56	73.98 $\pm$ 6.28	74.50 $\pm$ 6.42	72.72 $\pm$ 6.48	74.89 $\pm$ 6.26
	AUPR	75.85	68.85	69.63	75.78	72.62	OOM	79.99 $\pm$ 12.80	80.67 $\pm$ 12.37	80.15	80.47	82.50 $\pm$ 11.31	81.55 $\pm$ 9.37	81.62 $\pm$ 11.86	83.25 $\pm$ 10.80
	FPR95	90.59	100.0	94.24	90.80	93.84	OOM	87.90 $\pm$ 2.57	86.45 $\pm$ 2.97	85.16	80.59	79.99 $\pm$ 4.62	77.53 $\pm$ 5.10	83.65 $\pm$ 4.14	77.37 $\pm$ 5.49
	ID ACC	53.78	51.39	51.59	53.36	50.76	OOM	53.21	53.10	52.39	53.26	53.28	51.32	52.44	52.41

TIDE: Graph Out-of-Distribution Detection via Tri-Component Information Decomposition

Table 19. **Cora**: Extended OOD detection performance with three types of OOD (Structure manipulation, Feature interpolation, and Label-leave-out). GS++ is short for GNNSAFE++ and NS++ is short for NODESAFE++. OE indicates OOD exposure.

Dataset	Metrics	Non-OOD Exposure								Real OOD Exposure				TIDE	
		MSP	ODIN	Mahalanobis	Energy	GKDE	GNP	GNNSAFE	NODESAFE	OE	Energy FT	GS++	NS++	w/o OE	w/ OE
Cora-S	AUROC	70.90	49.92	46.68	71.73	68.61	77.47	87.64 ± 0.39	92.27 ± 0.74	67.98	75.88	90.19 ± 0.46	89.57 ± 8.12	95.15 ± 0.37	95.31 ± 0.37
	AUPR	45.73	27.01	29.03	46.08	44.26	53.26	77.93 ± 0.59	82.32 ± 0.61	46.93	49.18	81.37 ± 1.02	80.86 ± 9.20	89.33 ± 0.57	89.44 ± 0.41
	FPR95	87.30	100.00	98.19	88.74	84.34	76.22	73.49 ± 1.91	38.06 ± 4.49	95.31	67.73	56.81 ± 0.81	45.11 ± 32.56	23.31 ± 1.04	22.62 ± 3.11
	ID ACC	75.50	74.90	74.90	76.00	73.70	76.50	77.53 ± 0.38	78.97 ± 0.32	71.80	75.50	77.67 ± 0.42	67.67 ± 12.64	78.97 ± 1.22	77.90 ± 0.46
Cora-F	AUROC	85.39	49.88	49.93	86.15	82.79	85.88	92.76 ± 0.43	96.11 ± 0.14	81.83	88.15	95.22 ± 0.58	92.52 ± 4.91	97.88 ± 0.24	98.23 ± 0.14
	AUPR	73.70	26.96	31.95	74.42	66.52	73.79	87.85 ± 1.07	91.87 ± 0.75	70.84	75.99	90.27 ± 1.26	88.00 ± 4.94	94.67 ± 0.56	95.74 ± 0.48
	FPR95	64.88	100.00	99.93	65.81	68.24	56.17	48.56 ± 4.52	15.50 ± 1.48	83.79	47.53	29.01 ± 3.95	46.51 ± 37.56	8.13 ± 1.32	7.08 ± 0.85
	ID ACC	75.30	75.00	74.90	76.10	74.80	77.00	77.50 ± 0.26	78.73 ± 0.40	73.30	75.30	77.67 ± 0.51	66.23 ± 8.94	78.10 ± 0.89	78.67 ± 0.70
Cora-L	AUROC	91.36	49.80	67.62	91.40	57.23	90.34	93.19 ± 0.11	91.68 ± 1.09	89.47	91.36	93.99 ± 0.79	91.86 ± 0.54	93.70 ± 0.04	93.73 ± 0.11
	AUPR	78.03	24.27	42.31	78.14	27.50	77.40	82.99 ± 0.45	79.73 ± 2.31	77.01	78.49	84.19 ± 2.05	79.37 ± 0.16	84.03 ± 0.08	83.86 ± 0.23
	FPR95	34.99	100.00	90.77	41.08	88.95	37.42	29.55 ± 2.24	34.14 ± 6.75	46.55	37.83	26.03 ± 3.76	35.80 ± 8.21	28.84 ± 0.36	28.23 ± 0.26
	ID ACC	88.92	88.92	88.92	88.92	89.87	91.46	89.66 ± 0.66	90.29 ± 0.48	87.97	90.51	91.14 ± 0.32	90.93 ± 0.80	90.61 ± 0.36	90.03 ± 0.41

Table 20. **Citeseer** Extended OOD detection performance with three types of OOD (Structure manipulation, Feature interpolation, and Label-leave-out). GS++ is short for GNNSAFE++ and NS++ is short for NODESAFE++. OE indicates OOD exposure.

Dataset	Metrics	Non-OOD Exposure								Real OOD Exposure				TIDE	
		MSP	ODIN	Mahalanobis	Energy	GKDE	GNP	GNNSAFE	NODESAFE	OE	Energy FT	GS++	NS++	w/o OE	w/ OE
Citeseer-S	AUROC	66.34	49.23	45.26	65.62	61.48	70.55	79.87 ± 0.56	87.70 ± 2.27	58.74	68.87	81.30 ± 0.43	77.32 ± 8.74	91.89 ± 0.36	91.90 ± 0.12
	AUPR	34.78	23.07	21.20	33.63	31.55	41.12	61.44 ± 1.50	76.81 ± 1.13	30.07	36.01	64.59 ± 0.73	59.74 ± 6.25	80.11 ± 0.89	79.86 ± 0.96
	FPR95	85.03	100.00	99.13	87.59	93.71	78.26	74.45 ± 0.59	65.99 ± 12.52	95.37	76.44	71.33 ± 2.10	72.31 ± 10.55	38.41 ± 2.29	37.09 ± 3.07
	ID ACC	65.60	66.10	60.70	65.20	64.70	65.80	65.20 ± 0.50	69.47 ± 0.85	59.00	63.00	64.93 ± 1.29	52.17 ± 11.42	70.03 ± 0.78	69.63 ± 0.51
Citeseer-F	AUROC	78.32	49.86	49.92	79.19	74.69	78.46	84.07 ± 0.41	85.11 ± 11.68	72.06	79.23	84.38 ± 0.31	84.12 ± 0.41	93.96 ± 0.73	93.79 ± 0.16
	AUPR	55.48	23.11	31.20	55.94	50.25	53.21	68.22 ± 0.73	75.22 ± 11.74	48.80	55.69	68.77 ± 0.84	68.84 ± 0.76	84.67 ± 0.35	83.61 ± 0.42
	FPR95	71.27	100.00	99.73	69.67	71.22	73.14	67.75 ± 1.69	60.72 ± 34.01	81.09	64.08	64.64 ± 0.32	65.69 ± 3.13	23.13 ± 3.55	23.32 ± 1.02
	ID ACC	66.20	65.80	53.30	64.50	64.20	63.20	64.70 ± 0.44	68.73 ± 0.47	60.50	64.40	65.03 ± 1.25	64.97 ± 1.35	68.40 ± 1.93	69.33 ± 0.21
Citeseer-L	AUROC	88.42	51.33	53.46	89.98	82.69	85.65	90.73 ± 0.17	90.59 ± 0.69	89.44	90.34	91.37 ± 0.29	91.44 ± 0.34	90.97 ± 0.23	90.77 ± 0.15
	AUPR	64.03	17.97	35.47	64.10	61.21	62.32	64.93 ± 0.58	65.01 ± 1.71	62.74	66.66	64.32 ± 1.55	66.00 ± 0.49	64.92 ± 1.49	64.24 ± 1.75
	FPR95	51.97	100.00	86.32	38.76	50.61	41.37	40.36 ± 1.19	33.55 ± 5.29	45.99	31.60	28.32 ± 0.95	30.35 ± 2.55	30.84 ± 0.90	27.24 ± 2.76
	ID ACC	89.36	89.36	72.51	90.58	89.16	89.30	89.46 ± 0.76	89.97 ± 0.30	87.23	90.58	88.25 ± 0.17	88.66 ± 0.50	88.55 ± 0.63	88.45 ± 0.34

Table 21. **Pubmed** Extended OOD detection performance with two types of OOD (Structure manipulation, Feature interpolation). GS++ is short for GNNSAFE++ and NS++ is short for NODESAFE++. Label-leave-out was left out as discussed in L. OE indicates OOD exposure.

Dataset	Metrics	Non-OOD Exposure								Real OOD Exposure				TIDE	
		MSP	ODIN	Mahalanobis	Energy	GKDE	GNP	GNNSAFE	NODESAFE	OE	Energy FT	GS++	NS++	w/o OE	w/ OE
Pubmed-S	AUROC	74.31	49.76	55.28	74.33	74.02	74.96	92.45 ± 1.15	93.17 ± 1.15	74.41	73.54	92.84 ± 0.28	95.32 ± 0.17	97.25 ± 0.58	98.46 ± 0.10
	AUPR	17.44	4.83	8.38	17.32	16.89	17.54	65.47 ± 2.38	69.31 ± 4.81	16.74	18.00	70.41 ± 0.33	68.07 ± 0.81	79.76 ± 2.17	85.51 ± 0.38
	FPR95	84.08	100.00	97.59	78.90	81.52	80.33	47.81 ± 6.10	49.54 ± 15.93	83.52	92.04	47.67 ± 6.87	20.32 ± 1.83	12.97 ± 3.87	7.55 ± 0.57
	ID ACC	75.10	75.30	69.30	75.60	75.20	75.80	77.00 ± 0.44	77.30 ± 0.62	72.90	75.80	77.63 ± 0.31	74.20 ± 0.10	75.93 ± 0.40	76.80 ± 0.52
Pubmed-F	AUROC	83.28	49.67	69.12	84.16	82.25	82.56	95.18 ± 0.13	89.00 ± 11.30	82.34	78.94	94.70 ± 0.51	94.96 ± 0.50	97.44 ± 0.59	98.05 ± 0.25
	AUPR	39.29	4.83	15.09	39.10	32.41	39.75	74.41 ± 1.56	67.81 ± 19.39	38.60	37.21	77.71 ± 0.31	76.34 ± 2.06	81.70 ± 2.92	85.21 ± 0.85
	FPR95	69.38	100.00	84.93	62.47	68.56	61.79	27.62 ± 2.55	50.80 ± 41.88	74.58	90.00	31.49 ± 5.20	29.43 ± 2.90	12.53 ± 1.63	10.12 ± 2.35
	ID ACC	75.00	75.30	73.00	75.50	74.10	74.50	76.57 ± 0.35	77.33 ± 1.04	73.10	75.30	78.13 ± 0.06	74.13 ± 1.16	76.40 ± 0.60	76.53 ± 0.38

Table 22. **Amazon-Photo**: Extended OOD detection performance with three types of OOD (Structure manipulation, Feature interpolation, and Label-leave-out).

Dataset	Metrics	Non-OOD Exposure								TIDE	
		MSP	ODIN	Mahalanobis	Energy	GKDE	GNP	GNNSAFE	NODESAFE		
Amazon-S	AUROC	98.27	93.24	71.69	98.51	76.39	97.17	98.60 ± 0.09	98.54 ± 0.05	98.97 ± 0.14	
	AUPR	98.54	95.26	79.01	98.72	81.58	96.39	99.22 ± 0.05	99.18 ± 0.03	99.42 ± 0.08	
	FPR95	6.13	65.44	99.91	4.97	99.25	11.65	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	
	ID ACC	92.84	92.84	92.79	92.86	87.57	88.51	92.64 ± 0.44	91.36 ± 0.02	92.95 ± 0.38	
Amazon-F	AUROC	97.31	81.15	76.50	97.87	58.96	87.91	98.46 ± 0.01	98.42 ± 0.03	98.65 ± 0.10	
	AUPR	95.16	78.47	71.14	95.64	66.76	84.77	98.90 ± 0.15	98.77 ± 0.17	99.04 ± 0.22	
	FPR95	8.72	100.0	76.12	6.00	99.28	49.11	0.44 ± 0.28	0.30 ± 0.05	0.30 ± 0.06	
	ID ACC	92.89	92.71	92.86	92.96	86.18	90.05	92.65 ± 0.55	91.55 ± 0.41	92.70 ± 0.33	
Amazon-L	AUROC	93.97	65.97	73.25	93.81	65.58	92.72	96.92 ± 0.33	96.56 ± 0.35	96.86 ± 0.35	
	AUPR	91.32	57.80	66.89	91.13	65.20	90.34	96.38 ± 0.70	95.35 ± 0.58	95.79 ± 0.39	
	FPR95	26.65	90.23	74.30	28.48	96.87	37.16	9.29 ± 1.15	10.78 ± 0.99	8.13 ± 3.55	
	ID ACC	95.76	96.08	95.76	95.72	89.37	90.07	96.10 ± 0.29	95.20 ± 0.11	96.07 ± 0.20	

TIDE: Graph Out-of-Distribution Detection via Tri-Component Information Decomposition

Table 23. Coauthor: Extended OOD detection performance with three types of OOD (Structure manipulation, Feature interpolation, and Label-leave-out).

Dataset	Metrics	Non-OOD Exposure								TIDE
		MSP	ODIN	Mahalanobis	Energy	GKDE	GPN	GNNSAFE	NODESAFE	
Coauthor-S	AUROC	95.30	52.14	80.46	96.18	65.87	34.67	99.80 ± 0.17	99.85 ± 0.07	99.95 ± 0.02
	AUPR	94.37	48.83	76.65	95.25	72.65	40.21	99.82 ± 0.11	99.85 ± 0.06	99.94 ± 0.02
	FPR95	24.75	99.92	70.75	18.02	99.48	99.57	0.26 ± 0.02	0.23 ± 0.06	0.13 ± 0.03
	ID ACC	92.47	92.34	92.33	92.75	88.62	89.45	92.81 ± 0.10	92.11 ± 0.26	92.72 ± 0.11
Coauthor-F	AUROC	97.05	51.54	93.23	97.88	80.69	81.77	99.80 ± 0.14	99.86 ± 0.07	99.94 ± 0.02
	AUPR	96.93	45.50	90.88	97.69	86.47	80.56	99.78 ± 0.11	99.82 ± 0.07	99.92 ± 0.04
	FPR95	15.55	100.0	28.10	9.75	96.57	74.46	0.39 ± 0.11	0.30 ± 0.03	0.19 ± 0.05
	ID ACC	92.45	92.39	92.34	92.75	84.72	87.05	95.38 ± 0.16	95.35 ± 0.15	95.47 ± 0.07
Coauthor-L	AUROC	94.88	51.44	85.36	95.87	61.15	93.24	97.35 ± 0.26	97.34 ± 0.27	97.92 ± 0.04
	AUPR	97.99	74.79	93.61	98.34	81.39	97.55	99.03 ± 0.10	99.03 ± 0.10	99.24 ± 0.02
	FPR95	23.81	100.0	45.41	18.69	94.60	34.78	12.22 ± 1.09	12.46 ± 1.22	9.60 ± 0.60
	ID ACC	95.18	95.15	95.19	95.20	89.05	91.68	95.38 ± 0.16	95.35 ± 0.15	95.47 ± 0.07

Table 24. Twitch: Extended OOD detection performance on OOD Twitch sub-graphs ES, FR and RU. GS++ is short for GNNSAFE++ and NS++ is short for NODESAFE++. OE indicates OOD exposure.

Dataset	Metrics	Non-OOD Exposure								Real OOD Exposure				TIDE	
		MSP	ODIN	Mahalanobis	Energy	GKDE	GPN	GNNSAFE	NODESAFE	OE	Energy FT	GS++	NS++	w/o OE	w/ OE
Twitch-ES	AUROC	37.72	83.83	45.66	38.80	48.70	53.00	70.20 ± 18.91	70.66 ± 18.90	55.97	80.73	94.53 ± 2.47	95.48 ± 1.20	95.97 ± 1.31	96.80 ± 1.33
	AUPR	53.08	80.43	58.82	54.26	61.05	64.24	75.88 ± 17.90	76.13 ± 17.89	69.49	87.56	97.14 ± 1.33	97.42 ± 0.47	96.30 ± 0.57	97.10 ± 2.47
	FPR95	98.09	33.28	95.48	95.70	95.37	95.05	90.48 ± 4.55	89.56 ± 5.20	94.94	76.76	40.36 ± 22.32	27.99 ± 17.57	17.70 ± 13.41	7.38 ± 7.06
	ID ACC	68.72	70.79	70.51	70.40	67.44	68.09	70.75 ± 0.69	70.75 ± 0.69	70.73	70.52	70.36 ± 0.30	70.07 ± 0.33	68.63 ± 1.41	68.74 ± 0.79
Twitch-FR	AUROC	21.82	59.82	40.40	57.21	49.19	51.25	49.44 ± 28.72	49.38 ± 29.23	45.66	79.66	93.91 ± 1.73	37.28 ± 52.74	86.80 ± 12.59	84.22 ± 16.97
	AUPR	38.27	64.63	46.69	61.48	52.94	55.37	57.81 ± 20.56	57.90 ± 20.87	54.03	81.20	95.94 ± 0.99	56.13 ± 36.96	89.05 ± 11.22	88.71 ± 12.74
	FPR95	99.25	92.57	95.54	91.57	95.04	93.92	94.69 ± 7.17	94.62 ± 7.25	95.48	76.39	45.82 ± 13.57	67.09 ± 55.37	52.43 ± 40.23	65.89 ± 27.51
	ID ACC	68.72	70.79	70.51	70.40	67.44	68.09	70.75 ± 0.69	70.75 ± 0.69	70.73	70.52	70.36 ± 0.30	70.07 ± 0.33	68.63 ± 1.41	68.74 ± 0.79
Twitch-RU	AUROC	41.23	58.67	55.68	57.72	46.48	50.89	79.34 ± 16.34	79.39 ± 16.59	55.72	93.12	98.79 ± 0.92	97.60 ± 0.57	86.38 ± 0.77	93.53 ± 0.79
	AUPR	56.06	72.58	66.42	66.68	62.11	65.14	84.07 ± 9.53	84.09 ± 9.73	70.18	95.36	99.28 ± 0.59	98.36 ± 0.64	90.41 ± 2.50	94.95 ± 4.94
	FPR95	95.01	93.98	90.13	87.57	95.62	99.93	57.37 ± 34.19	57.67 ± 34.76	95.07	30.72	3.25 ± 3.97	8.29 ± 5.27	69.67 ± 7.40	26.29 ± 18.01
	ID ACC	68.72	70.79	70.51	70.40	67.44	68.09	70.75 ± 0.69	70.75 ± 0.69	70.73	70.52	70.36 ± 0.30	70.07 ± 0.33	68.63 ± 1.41	68.74 ± 0.79

Table 25. Arxiv: Extended OOD detection performance on OOD dataset of papers published in 2018, 2019, and 2020. GS++ is short for GNNSAFE++ and NS++ is short for NODESAFE++. OE indicates OOD exposure.

Dataset	Metrics	Non-OOD Exposure								Real OOD Exposure				TIDE	
		MSP	ODIN	Mahalanobis	Energy	GKDE	GPN	GNNSAFE	NODESAFE	OE	Energy FT	GS++	NS++	w/o OE	w/ OE
Arxiv-2018	AUROC	61.66	53.49	57.08	61.75	56.29	OOM	65.94 ± 0.38	66.73 ± 0.16	67.72	69.58	69.47 ± 0.63	69.81 ± 0.58	67.97 ± 0.10	70.5 ± 0.20
	AUPR	70.63	63.06	65.09	70.41	66.78	OOM	74.37 ± 0.29	75.20 ± 0.27	75.74	76.31	77.63 ± 0.64	77.68 ± 0.48	76.42 ± 0.20	78.59 ± 0.21
	FPR95	91.67	100.0	93.69	91.74	94.31	OOM	89.88 ± 0.67	88.80 ± 0.37	86.67	82.10	83.43 ± 1.24	81.51 ± 1.19	86.88 ± 0.48	81.57 ± 0.25
	ID ACC	53.78	51.39	51.59	53.36	50.76	OOM	53.21 ± 0.16	53.10 ± 0.79	52.39	53.26	53.28 ± 0.35	51.32 ± 0.21	52.44 ± 0.24	52.41 ± 0.29
Arxiv-2019	AUROC	63.07	53.95	56.76	63.16	57.87	OOM	67.90 ± 0.37	68.76 ± 0.16	69.33	70.58	71.32 ± .63	71.87 ± 0.59	70.09 ± 0.21	72.30 ± 0.23
	AUPR	66.00	56.07	57.85	65.78	62.34	OOM	70.97 ± 0.29	71.98 ± .33	72.15	72.03	74.45 ± 0.77	74.74 ± 0.68	73.24 ± 0.24	75.56 ± 0.30
	FPR95	90.82	100.0	94.01	90.96	93.97	OOM	88.83 ± 0.72	87.44 ± 0.23	85.52	81.30	81.80 ± 1.37	79.31 ± 1.13	85.10 ± 0.75	79.39 ± 0.38
	ID ACC	53.78	51.39	51.59	53.36	50.76	OOM	53.21 ± 0.16	53.10 ± 0.79	52.39	53.26	53.28 ± 0.35	51.32 ± 0.21	52.44 ± 0.24	52.41 ± 0.29
Arxiv-2020	AUROC	67.00	55.78	56.92	67.70	60.79	OOM	77.90 ± 0.32	78.59 ± 0.10	72.35	74.53	81.15 ± 0.51	81.82 ± 0.50	80.10 ± 0.23	82.03 ± 0.23
	AUPR	90.92	87.41	85.95	91.15	88.74	OOM	94.64 ± 0.08	94.83 ± 0.03	92.57	93.08	95.43 ± .12	92.23 ± 5.66	95.19 ± 0.07	95.59 ± 0.06
	FPR95	89.28	100.0	95.01	89.69	93.31	OOM	85.00 ± 0.98	83.11 ± 0.42	83.28	78.36	74.75 ± 1.99	71.78 ± 1.32	78.98 ± 0.56	71.15 ± 0.62
	ID ACC	53.78	51.39	51.59	53.36	50.76	OOM	53.21 ± 0.16	53.10 ± 0.79	52.39	53.26	53.28 ± 0.35	51.32 ± 0.21	52.44 ± 0.24	52.41 ± 0.29

Table 26. Cora: Extended ablation performance of TIDE.

Model	Cora-S				Cora-F				Cora-L			
	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc
GNNSAFE	87.64 ± 0.39	77.93 ± 0.59	73.49 ± 1.91	77.53 ± 0.38	92.76 ± 0.43	87.85 ± 1.07	48.56 ± 4.52	77.50 ± 0.26	93.19 ± 0.11	82.99 ± 0.45	29.55 ± 2.24	89.66 ± 0.66
$\mathcal{L}_{VIB}$	94.79 ± 0.22	88.41 ± 0.49	26.22 ± 2.55	78.73 ± 0.50	97.69 ± 0.13	93.94 ± 0.36	9.45 ± 0.78	77.43 ± 0.58	93.06 ± 0.11	83.12 ± 0.11	30.93 ± 0.57	90.30 ± 0.37
$\mathcal{L}_{VIB} \& \mathcal{L}_{VCI_{ind}}$	94.86 ± 0.29	88.48 ± 0.54	24.84 ± 0.65	78.60 ± 0.40	97.72 ± 0.10	94.26 ± 0.52	9.40 ± 1.06	77.47 ± 0.45	93.11 ± 0.16	83.20 ± 0.18	30.43 ± 0.11	90.30 ± 0.18
TIDE	95.15 ± 0.37	89.33 ± 0.57	23.31 ± 1.04	78.97 ± 1.22	97.88 ± 0.24	94.67 ± 0.56	8.13 ± 1.32	78.10 ± 0.89	93.70 ± 0.04	84.03 ± 0.08	28.84 ± 0.36	90.61 ± 0.36

Table 27. **Citeseer**: Extended ablation performance of TIDE.

Model	Citeseer-S				Citeseer-F				Citeseer-L			
	AUROC	AUPR	FPR	ID Acc	AUROC	AUPR	FPR	ID Acc	AUROC	AUPR	FPR	ID Acc
GNNSAFE	79.87 $\pm$ 0.56	61.44 $\pm$ 1.50	74.45 $\pm$ 0.59	65.20 $\pm$ 0.50	84.07 $\pm$ 0.41	68.22 $\pm$ 0.73	67.75 $\pm$ 1.69	64.70 $\pm$ 0.44	90.73 $\pm$ 0.17	64.93 $\pm$ 0.58	40.36 $\pm$ 1.19	89.46 $\pm$ 0.76
$\mathcal{L}_{VIB}$	90.84 $\pm$ 0.57	78.76 $\pm$ 0.47	45.62 $\pm$ 3.06	69.17 $\pm$ 0.49	93.01 $\pm$ 0.64	82.83 $\pm$ 1.64	31.77 $\pm$ 5.79	69.27 $\pm$ 0.49	90.66 $\pm$ 0.10	64.06 $\pm$ 0.67	34.87 $\pm$ 3.12	87.74 $\pm$ 0.35
$\mathcal{L}_{VIB} \& \mathcal{L}_{VCInd}$	91.85 $\pm$ 0.31	80.29 $\pm$ 0.69	39.61 $\pm$ 2.68	69.13 $\pm$ 0.21	93.25 $\pm$ 0.87	82.94 $\pm$ 2.15	26.96 $\pm$ 4.64	67.83 $\pm$ 1.42	90.88 $\pm$ 0.25	64.78 $\pm$ 1.62	31.14 $\pm$ 1.19	88.75 $\pm$ 0.31
TIDE	91.89 $\pm$ 0.36	80.11 $\pm$ 0.89	38.41 $\pm$ 2.29	70.03 $\pm$ 0.78	93.96 $\pm$ 0.73	84.67 $\pm$ 2.35	23.13 $\pm$ 3.55	68.40 $\pm$ 1.93	90.97 $\pm$ 0.23	64.92 $\pm$ 1.49	30.84 $\pm$ 0.90	88.55 $\pm$ 0.63

Table 28. **Pubmed**: Extended ablation performance of TIDE.

Model	Pubmed-S				Pubmed-F			
	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc
GNNSAFE	92.45 $\pm$ 1.15	65.47 $\pm$ 2.38	47.81 $\pm$ 6.10	77.00 $\pm$ 0.44	95.18 $\pm$ 0.13	74.41 $\pm$ 1.56	27.62 $\pm$ 2.55	76.57 $\pm$ 0.35
$\mathcal{L}_{VIB}$	96.18 $\pm$ 0.44	75.74 $\pm$ 1.60	17.64 $\pm$ 3.69	74.60 $\pm$ 1.71	96.98 $\pm$ 0.41	78.94 $\pm$ 3.33	16.05 $\pm$ 1.64	75.30 $\pm$ 1.22
$\mathcal{L}_{VIB} \& \mathcal{L}_{VCInd}$	96.45 $\pm$ 0.64	76.46 $\pm$ 2.31	15.66 $\pm$ 4.88	74.77 $\pm$ 1.86	97.42 $\pm$ 0.51	81.07 $\pm$ 3.23	13.08 $\pm$ 2.41	75.60 $\pm$ 0.46
TIDE	97.25 $\pm$ 0.58	79.76 $\pm$ 2.17	12.97 $\pm$ 3.87	75.93 $\pm$ 0.40	97.44 $\pm$ 0.59	81.70 $\pm$ 2.92	12.53 $\pm$ 1.63	76.40 $\pm$ 0.60

Table 29. **Twitch**: Extended ablation performance of TIDE.

Model	Twitch-ES				Twitch-FR				Twitch-RU			
	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc
GNNSAFE	70.20 $\pm$ 18.91	75.88 $\pm$ 17.90	90.48 $\pm$ 4.55	70.75 $\pm$ 0.69	49.44 $\pm$ 28.72	57.81 $\pm$ 20.56	94.69 $\pm$ 7.17	70.75 $\pm$ 0.69	79.34 $\pm$ 16.34	84.07 $\pm$ 9.53	58.37 $\pm$ 34.19	70.75 $\pm$ 0.69
$\mathcal{L}_{VIB}$	83.29 $\pm$ 11.53	86.36 $\pm$ 10.78	56.60 $\pm$ 33.50	70.36 $\pm$ 0.35	57.97 $\pm$ 14.37	58.80 $\pm$ 12.38	88.70 $\pm$ 13.46	70.36 $\pm$ 0.35	79.23 $\pm$ 13.76	84.86 $\pm$ 8.38	62.48 $\pm$ 36.82	70.36 $\pm$ 0.35
$\mathcal{L}_{VIB} \& \mathcal{L}_{VCInd}$	95.04 $\pm$ 0.11	96.05 $\pm$ 0.13	19.35 $\pm$ 2.60	70.33 $\pm$ 0.63	75.14 $\pm$ 1.03	80.85 $\pm$ 0.58	90.69 $\pm$ 5.46	70.33 $\pm$ 0.63	92.67 $\pm$ 2.40	94.93 $\pm$ 0.93	39.68 $\pm$ 23.80	70.33 $\pm$ 0.63
TIDE	95.97 $\pm$ 1.31	96.30 $\pm$ 0.57	17.70 $\pm$ 13.41	68.63 $\pm$ 1.41	86.80 $\pm$ 12.59	89.05 $\pm$ 11.22	52.43 $\pm$ 40.23	68.63 $\pm$ 1.41	86.38 $\pm$ 0.77	90.41 $\pm$ 2.50	69.67 $\pm$ 7.40	68.63 $\pm$ 1.41

Table 30. **ArXiv**: Extended ablation performance of TIDE.

Model	ArXiv-18				ArXiv-19				ArXiv-20			
	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc	AUROC	AUPR	FPR95	ID Acc
GNNSAFE	65.94 $\pm$ 0.38	74.37 $\pm$ 0.29	89.88 $\pm$ 0.67	53.21 $\pm$ 0.16	67.90 $\pm$ 0.37	70.97 $\pm$ 0.29	88.83 $\pm$ 0.72	53.21 $\pm$ 0.16	77.90 $\pm$ 0.32	94.64 $\pm$ 0.08	85.00 $\pm$ 0.98	53.21 $\pm$ 0.16
$\mathcal{L}_{VIB}$	67.41 $\pm$ 0.95	75.84 $\pm$ 1.02	87.72 $\pm$ 1.28	52.46 $\pm$ 0.24	69.49 $\pm$ 1.08	72.51 $\pm$ 1.09	86.35 $\pm$ 1.35	52.46 $\pm$ 0.24	79.53 $\pm$ 1.04	95.05 $\pm$ 0.27	80.72 $\pm$ 1.85	52.46 $\pm$ 0.24
$\mathcal{L}_{VIB} \& \mathcal{L}_{VCInd}$	67.95 $\pm$ 0.11	76.40 $\pm$ 0.21	86.97 $\pm$ 0.46	52.43 $\pm$ 0.22	70.03 $\pm$ 0.25	73.20 $\pm$ 0.23	85.43 $\pm$ 0.89	52.43 $\pm$ 0.22	80.07 $\pm$ 0.25	95.12 $\pm$ 0.17	79.18 $\pm$ 0.65	52.43 $\pm$ 0.22
TIDE	67.97 $\pm$ 0.10	76.42 $\pm$ 0.20	86.88 $\pm$ 0.48	52.44 $\pm$ 0.24	70.09 $\pm$ 0.21	73.24 $\pm$ 0.24	85.10 $\pm$ 0.75	52.44 $\pm$ 0.24	80.10 $\pm$ 0.23	95.19 $\pm$ 0.07	78.98 $\pm$ 0.56	52.44 $\pm$ 0.24

N. Computational Cost

Table 31 presents a comparison of the computational cost of TIDE against GNNSafe and NODESAFE, all evaluated using the same backbone. Evidently, **TIDE consistently outperforms the baselines, achieving superior OOD detection performance at the cost of a 2-3x increase in training time, a marginal rise in memory usage, and a comparable inference time.** Despite the additional training cost, this **trade-off is deemed acceptable due to the significantly improved detection performance and competitive inference time**, as discussed in Appendix B. These results highlight the strong performance and practical applicability of TIDE for node-level graph OOD detection.

Table 31. **Computation cost (single 48GB (49140MiB) NVIDIA RTX A6000 GPU) and OOD detection performance of TIDE against SOTA Non-OOD exposure baselines.** The ‘Train’ column is the training convergence time in seconds. The ‘In.’ column is the inference time in seconds. The ‘Mem.’ column is the maximum memory usage in Megabytes (MiB). The ‘FPR95’ column is the OOD detection performance in %, the lower the better.

	Cora - Structure				Citeseer - Structure				Pubmed - Structure				Twitch				Arxiv			
	Train	In.	Mem.	FPR95(↓)	Train	In.	Mem.	FPR95(↓)	Train	In.	Mem.	FPR95(↓)	Train	In.	Mem.	FPR95(↓)	Train	In.	Mem.	FPR95(↓)
GNNSAFE	2.05	0.01	625	73.49	2.25	0.02	693	74.45	4.22	0.02	827	47.81	3.55	0.03	749	81.18	19.72	0.11	3055	87.90
NODESAFE	0.76	0.02	625	38.06	1.45	0.02	693	65.99	1.69	0.02	827	49.54	1.35	0.03	749	80.62	16.79	0.11	3055	86.45
TIDE	4.02	0.02	653	23.31	9.85	0.02	780	38.41	9.96	0.024	906	12.97	8.01	0.028	794	43.14	39.7	0.13	3490	83.65